

Coconut Water ML Models — Full Documentation

Complete model type details, full source code (black panels only), and full training datasets on a white page layout with wide left/right margins and spacing between sections.

System overview

Project	Ceylon coconut-water authenticity / composition prediction
ML libraries	scikit-learn, pandas, numpy, joblib
Model family	RandomForestRegressor (ensemble of decision trees)
Modules	ML_1 (acids) + ML_2 (sugar)
Train / test split	80% train / 20% test, random_state=42
Authenticity	Rule-based natural-range check (not a classifier)

1. ML_1 — Acid model (citric & ascorbic)

Model type & configuration

Model type	Multi-output RandomForestRegressor (scikit-learn)
Saved artifact	ML_1/models/random_forest_ph_temp.pkl
Hyperparameters	n_estimators=300, max_depth=12, min_samples_leaf=2, random_state=42
Input features	pH, temperature_C
Output targets	citric_percent_wv, ascorbic_percent_wv
Training CSV	ML_1/dataset/combined_pH_temp_600.csv (600 rows)
Data origin	Literature maturation curves → synthetic combined dataset

Full source code — ML_1

Code appears only inside black panels below. Surrounding page, margins, and headings stay white.

ML_1/load_dataset.py

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ML_1/load_dataset.py

"""
Phase 3.1 ? Load combined dataset: pH, temperature -> citric_percent_wv, ascorbic_percent_wv.
"""

from pathlib import Path
from typing import Tuple

import numpy as np
import pandas as pd

FEATURE_COLUMNS = ["pH", "temperature_C"]
TARGET_COLUMNS = ["citric_percent_wv", "ascorbic_percent_wv"]

def get_dataset_path() -> Path:
    return Path(__file__).resolve().parent / "dataset" / "combined_pH_temp_600.csv"

def load_dataset(csv_path: str | Path | None = None) -> pd.DataFrame:
    path = Path(csv_path) if csv_path else get_dataset_path()
    if not path.exists():
        raise FileNotFoundError(f"Dataset not found: {path}. Run merge_datasets.py first.")
    df = pd.read_csv(path)
    df.columns = df.columns.str.strip()
    return df

def prepare_xy(df: pd.DataFrame) -> Tuple[np.ndarray, np.ndarray]:
    for col in FEATURE_COLUMNS + TARGET_COLUMNS:
        if col not in df.columns:
            raise ValueError(f"Missing column: {col}")
    df_clean = df[FEATURE_COLUMNS + TARGET_COLUMNS].dropna()
    X = df_clean[FEATURE_COLUMNS].astype(np.float64).values
    y = df_clean[TARGET_COLUMNS].astype(np.float64).values
    return X, y

def load_xy(csv_path: str | Path | None = None) -> Tuple[np.ndarray, np.ndarray]:
    df = load_dataset(csv_path)
    return prepare_xy(df)

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ML_1/merge_datasets.py

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ML_1/merge_datasets.py

"""
Merge citric and ascorbic datasets by nearest (pH, temperature) to create
combined training data: pH, temperature -> citric_percent_wv, ascorbic_percent_wv.

Run from project root: python ML_1/merge_datasets.py
"""

import numpy as np
import pandas as pd
from pathlib import Path

PROJECT_ROOT = Path(__file__).resolve().parent.parent
CITRIC_PATH = PROJECT_ROOT / "citric_pH_temp_synthetic_600.csv"
ASCORBIC_PATH = PROJECT_ROOT / "ascorbic_pH_temp_synthetic_600.csv"
OUT_PATH = Path(__file__).resolve().parent / "dataset" / "combined_pH_temp_600.csv"

def merge_by_nearest():
    citric = pd.read_csv(CITRIC_PATH)
    ascorbic = pd.read_csv(ASCORBIC_PATH)

    X_citric = citric[["pH", "temperature_C"]].values
    X_ascorbic = ascorbic[["pH", "temperature_C"]].values
    citric_pct = citric["citric_percent_wv"].values
    ascorbic_pct = ascorbic["ascorbic_percent_wv"].values

    # For each citric row, find nearest ascorbic row in (pH, temp) space
    merged_rows = []
    for i in range(len(X_citric)):
        dists = np.sqrt(
            (X_ascorbic[:, 0] - X_citric[i, 0]) ** 2
            + ((X_ascorbic[:, 1] - X_citric[i, 1]) / 10) ** 2 # scale temp
        )
        j = np.argmin(dists)
        merged_rows.append({
            "pH": X_citric[i, 0],
            "temperature_C": X_citric[i, 1],
            "citric_percent_wv": citric_pct[i],
            "ascorbic_percent_wv": ascorbic_pct[j],
        })

    df = pd.DataFrame(merged_rows)
    OUT_PATH.parent.mkdir(parents=True, exist_ok=True)
    df.to_csv(OUT_PATH, index=False)
    print(f"Saved {len(df)} rows to {OUT_PATH}")
    print(df.head())
    return df

if __name__ == "__main__":
    merge_by_nearest()

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ML_1/train_model.py

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ML_1/train_model.py

"""
Phase 3.2 ? Train Random Forest: pH, temperature -> citric_percent_wv, ascorbic_percent_wv.
"""

from pathlib import Path

import joblib
import numpy as np
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.model_selection import train_test_split

try:
    from load_dataset import TARGET_COLUMNS, load_xy
except ImportError:
    from ML_1.load_dataset import TARGET_COLUMNS, load_xy

RANDOM_STATE = 42
TEST_SIZE = 0.2
MODEL_PARAMS = {"n_estimators": 300, "max_depth": 12, "random_state": RANDOM_STATE, "min_samples_leaf": 2}

def get_models_dir() -> Path:
    return Path(__file__).resolve().parent / "models"

def train_and_evaluate(csv_path: str | Path | None = None) -> RandomForestRegressor:
    X, y = load_xy(csv_path)
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=TEST_SIZE, random_state=RANDOM_STATE
    )
    model = RandomForestRegressor(**MODEL_PARAMS)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    print("Evaluation (test set):")
    print("-" * 50)
    for i, name in enumerate(TARGET_COLUMNS):
        r2 = r2_score(y_test[:, i], y_pred[:, i])
        rmse = np.sqrt(mean_squared_error(y_test[:, i], y_pred[:, i]))
        mae = mean_absolute_error(y_test[:, i], y_pred[:, i])
        print(f" {name}: R2={r2:.4f}, RMSE={rmse:.4f}, MAE={mae:.4f}")
    r2_avg = r2_score(y_test, y_pred)
    rmse_avg = np.sqrt(mean_squared_error(y_test, y_pred))
    mae_avg = mean_absolute_error(y_test, y_pred)
    print("-" * 50)
    print(f" (average): R2={r2_avg:.4f}, RMSE={rmse_avg:.4f}, MAE={mae_avg:.4f}")
    return model

def save_model(model: RandomForestRegressor, path: str | Path | None = None) -> Path:
    out_dir = get_models_dir()
    out_dir.mkdir(parents=True, exist_ok=True)
    out_path = path or out_dir / "random_forest_ph_temp.pkl"
    joblib.dump(model, out_path)
    print(f"Model saved to {out_path}")

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ML_1/train_model.py (cont.)

    return Path(out_path)

if __name__ == "__main__":
    model = train_and_evaluate()
    save_model(model)

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ML_1/predict.py

"""
Phase 3.3 ? Prediction: pH, temperature -> citric_percent_wv, ascorbic_percent_wv.
"""

from pathlib import Path
from typing import Any

import joblib
import numpy as np

_MODEL: Any = None
_MODEL_PATH = Path(__file__).resolve().parent / "models" / "random_forest_ph_temp.pkl"

def _get_model() -> Any:
    global _MODEL
    if _MODEL is None:
        if not _MODEL_PATH.exists():
            raise FileNotFoundError(
                f"Model not found: {_MODEL_PATH}. Run: python ML_1/train_model.py"
            )
        _MODEL = joblib.load(_MODEL_PATH)
    return _MODEL

def predict(pH: float, temperature: float) -> dict[str, float]:
    """Predict citric_percent_wv and ascorbic_percent_wv from pH and temperature."""
    model = _get_model()
    X = np.array([[pH, temperature]], dtype=np.float64)
    y_pred = model.predict(X)[0]
    return {
        "pH": float(pH),
        "temperature_C": float(temperature),
        "citric_percent_wv": float(y_pred[0]),
        "ascorbic_percent_wv": float(y_pred[1]),
    }

```

ML_1 full training dataset

File: ML_1/dataset/combined_pH_temp_600.csv — all 600 training rows (white table surface).

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.564	30.02	0.07491	0.001041
5.789	29.0	0.06043	0.000895
5.239	27.01	0.10291	0.002206
5.062	18.89	0.11719	0.002622
5.565	31.59	0.07209	0.001173
5.028	18.61	0.11859	0.002757
5.564	22.56	0.07361	0.001324
4.995	24.66	0.11253	0.002725
5.303	29.65	0.09396	0.001985
5.152	19.96	0.11113	0.002454
5.643	27.87	0.07099	0.001154
5.404	19.6	0.08513	0.001799
5.398	26.89	0.0872	0.001738
5.12	24.11	0.10863	0.002353
5.631	18.82	0.07199	0.001165
5.485	28.97	0.08089	0.001413
5.615	18.32	0.07192	0.001109
5.289	25.01	0.09449	0.001985
5.287	22.22	0.09512	0.001877
4.821	31.47	0.12101	0.002806
5.144	28.9	0.10925	0.002346
5.061	30.64	0.1083	0.002249
5.146	20.47	0.1031	0.002403
5.566	26.78	0.08109	0.001295
4.877	18.58	0.13049	0.003106
5.003	26.23	0.11594	0.002707
5.371	26.27	0.08404	0.001529
5.343	19.16	0.09954	0.002173
5.711	24.63	0.06463	0.000961
5.345	18.3	0.09434	0.001861
4.897	19.52	0.11992	0.002865
5.569	19.51	0.07656	0.001239
4.782	23.19	0.12546	0.003185
5.186	22.07	0.10538	0.002349
5.767	18.63	0.06179	0.000951
5.773	30.47	0.06469	0.001079
5.156	27.26	0.1071	0.002342
5.589	31.12	0.06925	0.001359
5.649	20.51	0.07578	0.001085
5.167	31.61	0.10238	0.002268
4.839	19.85	0.13347	0.003062
5.322	26.14	0.08658	0.0018
5.499	19.83	0.07722	0.001334
5.359	31.38	0.08813	0.001776
4.806	26.88	0.1238	0.003114

Rows 1–45 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.362	26.34	0.09339	0.002026
5.29	18.25	0.09873	0.001929
5.54	25.43	0.06888	0.001149
5.257	27.13	0.09394	0.002041
5.021	28.44	0.11417	0.002704
4.833	20.05	0.13233	0.002994
4.974	20.32	0.11958	0.002839
4.899	18.78	0.12446	0.002904
5.476	23.51	0.08609	0.001645
5.666	18.61	0.06893	0.001296

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.432	18.69	0.08455	0.001817
4.849	19.39	0.13145	0.003159
5.768	28.91	0.06042	0.000936
4.922	25.2	0.12356	0.002616
5.231	21.73	0.09937	0.002036
5.328	23.15	0.09996	0.002055
5.175	22.16	0.10468	0.002235
5.376	25.5	0.09658	0.001931
5.621	26.86	0.06911	0.001255
5.047	30.05	0.11172	0.002674
4.956	19.18	0.12052	0.002997
5.748	26.31	0.06405	0.000938
5.759	26.26	0.06091	0.001035
5.435	24.52	0.08718	0.001703
5.693	26.39	0.06987	0.001232
4.969	27.21	0.12211	0.002801
5.554	30.58	0.07865	0.001429
5.195	20.21	0.10395	0.002325
5.277	28.22	0.09152	0.00192
5.645	23.27	0.07768	0.001291
5.784	29.91	0.06108	0.000828
5.415	18.17	0.09095	0.001555
5.061	24.19	0.10803	0.002432
5.556	31.96	0.07125	0.001141
5.69	31.99	0.06585	0.001125
4.886	18.41	0.12389	0.00301
5.641	30.75	0.07327	0.001169
4.805	26.48	0.12491	0.003047
5.645	29.21	0.06702	0.001127
5.589	18.56	0.07782	0.001518
5.439	20.17	0.07934	0.001497
5.13	31.72	0.10128	0.002412
5.478	30.38	0.07574	0.001611
4.857	31.72	0.1174	0.002898
5.491	20.03	0.0874	0.001525

Rows 46–90 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.466	27.09	0.08795	0.00168
5.372	24.64	0.09586	0.001838
5.149	23.99	0.10919	0.002435
5.22	21.05	0.09237	0.002023
4.798	24.64	0.13062	0.003027
5.396	27.07	0.0859	0.001728
5.66	24.04	0.07305	0.001418
5.419	30.41	0.08905	0.001776
5.338	22.4	0.08715	0.001873
5.681	21.93	0.0717	0.001129
5.072	18.21	0.10177	0.002278
5.245	26.41	0.09515	0.001933
5.865	28.81	0.05986	0.001022
5.531	26.97	0.0768	0.00141
5.4	20.32	0.09284	0.001954
5.367	31.86	0.08369	0.001781
5.29	28.57	0.09796	0.002224
4.914	30.7	0.11661	0.00259
5.1	26.21	0.09894	0.002316
5.034	26.93	0.10912	0.002478
5.604	23.13	0.0769	0.001297
5.69	28.76	0.07439	0.001113
5.695	29.9	0.06041	0.000967

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.252	26.83	0.0951	0.00217
4.996	30.98	0.11357	0.002615
5.44	21.89	0.08244	0.001649
5.842	24.69	0.06041	0.001002
5.511	28.95	0.07233	0.001357
5.521	25.74	0.07654	0.001417
5.801	28.27	0.05517	0.000853
5.588	31.8	0.07209	0.001042
5.23	19.19	0.09745	0.002077
5.769	18.52	0.06113	0.00111
4.91	27.99	0.12115	0.002795
5.604	18.69	0.0757	0.001424
5.681	25.2	0.0666	0.001157
5.225	27.4	0.09383	0.001893
5.285	22.19	0.0975	0.002246
5.095	20.63	0.11164	0.00252
5.583	31.98	0.07531	0.001264
4.959	31.97	0.11192	0.0026
5.216	24.96	0.0981	0.002264
4.872	18.25	0.12539	0.003107
5.165	31.43	0.09985	0.002127
5.11	21.29	0.10398	0.002485

Rows 91–135 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.275	26.27	0.0886	0.001713
5.098	20.18	0.11997	0.002606
5.615	24.14	0.07461	0.001304
5.497	20.59	0.0833	0.001576
5.477	25.18	0.08003	0.001558
5.737	18.58	0.05706	0.000947
5.293	22.38	0.09384	0.001905
5.265	27.77	0.09541	0.002171
5.259	18.02	0.10332	0.002102
4.758	24.85	0.12529	0.003253
5.571	21.64	0.07753	0.001047
5.291	21.2	0.09385	0.001993
4.78	27.96	0.1248	0.003072
4.948	18.13	0.12455	0.002955
4.812	21.38	0.12647	0.00285
5.676	29.95	0.07306	0.00115
5.569	21.56	0.07374	0.001158
5.519	23.06	0.08168	0.001683
5.646	21.03	0.0751	0.001269
4.875	22.85	0.12507	0.002962
5.629	28.34	0.07954	0.001312
5.439	20.01	0.08277	0.001689
5.305	18.06	0.09181	0.002027
5.43	30.56	0.08316	0.001594
5.458	18.71	0.08102	0.001638
5.432	18.25	0.08533	0.001581
5.696	26.55	0.0639	0.001122
5.568	25.94	0.07865	0.001319
5.27	18.17	0.09736	0.002136
5.224	30.46	0.09403	0.002063
4.905	20.5	0.12449	0.002739
5.242	30.3	0.09055	0.001878
5.47	28.96	0.07899	0.00116
5.69	29.17	0.06369	0.001162
5.566	31.71	0.07347	0.001372
5.711	22.47	0.07696	0.001164

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.289	31.1	0.09343	0.002108
5.226	19.84	0.1029	0.002248
5.2	21.41	0.09999	0.002295
5.673	20.83	0.06423	0.001078
5.313	18.63	0.09186	0.002039
5.418	19.55	0.08257	0.001499
5.597	30.62	0.07529	0.001116
5.72	19.93	0.06783	0.000992
4.817	26.63	0.11977	0.002877

Rows 136–180 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.066	29.93	0.10555	0.002373
5.064	23.67	0.10719	0.002381
4.997	19.69	0.11519	0.002574
5.048	20.51	0.11198	0.002551
5.109	29.92	0.11252	0.002506
4.998	20.26	0.10962	0.002562
5.602	24.87	0.07289	0.001251
5.281	19.76	0.09858	0.002156
5.776	25.76	0.05916	0.000996
4.837	30.84	0.12728	0.003082
5.323	25.45	0.09364	0.001703
5.499	24.56	0.07238	0.001208
5.147	22.79	0.10214	0.002462
5.602	20.97	0.07655	0.001189
5.479	31.91	0.07864	0.001415
5.305	20.13	0.09484	0.002015
5.696	28.17	0.06718	0.000998
5.461	31.3	0.07731	0.001588
5.532	18.56	0.07714	0.001486
5.609	25.44	0.07344	0.001318
5.831	28.34	0.06198	0.001028
5.561	30.7	0.07753	0.001288
5.16	24.16	0.10296	0.002295
5.492	27.28	0.08036	0.001336
4.99	30.05	0.10873	0.002511
4.914	19.94	0.12393	0.002922
5.742	25.59	0.0589	0.001026
5.455	26.76	0.08595	0.00167
5.22	18.33	0.09906	0.002167
5.431	31.1	0.08338	0.00147
5.42	26.85	0.08439	0.001661
5.75	29.81	0.0653	0.000935
5.602	24.73	0.07007	0.001351
5.503	21.61	0.08375	0.001511
5.018	30.05	0.10829	0.00253
5.188	29.75	0.10092	0.002217
4.899	31.98	0.11463	0.002814
5.377	31.06	0.09815	0.002021
4.953	21.79	0.12082	0.002817
4.842	20.47	0.12242	0.002891
5.383	27.14	0.08722	0.00193
4.903	31.67	0.12284	0.002946
5.291	27.41	0.0918	0.002082
5.625	23.83	0.07292	0.001276
5.584	30.5	0.07064	0.001168

Rows 181–225 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
4.87	25.47	0.11911	0.002961
4.736	28.84	0.12741	0.003128
5.645	20.06	0.06572	0.001114
5.359	19.3	0.08917	0.0019
5.085	25.06	0.10939	0.002618
5.125	20.92	0.10225	0.002284
4.856	20.45	0.12845	0.003028
5.807	27.15	0.06215	0.000962
5.754	30.22	0.06183	0.000982
5.487	28.29	0.07662	0.001387
5.726	18.37	0.07106	0.001094
5.33	23.86	0.09127	0.001766
5.413	21.79	0.08869	0.001619
5.378	18.93	0.08822	0.001627
5.727	20.41	0.06712	0.001219
5.313	18.83	0.09998	0.002116
4.97	21.0	0.11302	0.002589
5.282	19.89	0.0962	0.002039
5.162	23.0	0.10284	0.002301
5.012	31.02	0.10288	0.002452
4.818	19.71	0.12731	0.002934
5.663	31.95	0.06614	0.001176
5.133	26.08	0.10534	0.002254
5.739	24.91	0.05445	0.000983
5.206	19.79	0.10195	0.00221
4.944	26.25	0.12618	0.002854
5.277	30.97	0.08777	0.001966
5.201	25.95	0.10087	0.002309
5.376	22.84	0.08737	0.001873
5.011	23.45	0.11983	0.002716
5.585	29.58	0.07529	0.001273
5.57	31.24	0.07675	0.001438
5.425	18.54	0.09232	0.001732
4.846	29.93	0.11931	0.003128
4.782	22.63	0.13072	0.003375
5.418	20.25	0.08655	0.001556
5.418	31.49	0.08355	0.001624
5.441	23.24	0.0773	0.001684
5.32	31.75	0.086	0.001945
5.193	19.66	0.10496	0.002181
5.135	30.24	0.10439	0.00239
5.358	20.29	0.09297	0.001979
5.471	24.46	0.08393	0.001441
4.968	22.82	0.11664	0.002904
5.262	31.64	0.09641	0.002094

Rows 226–270 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.683	24.01	0.06927	0.001144
5.341	31.42	0.08665	0.001663
5.128	18.63	0.11024	0.002322
4.94	29.69	0.11974	0.002787
5.714	23.56	0.0643	0.00113
5.812	20.65	0.06408	0.000915
4.932	27.91	0.11649	0.002968
5.602	19.78	0.07202	0.001208
5.403	31.61	0.08876	0.00147
5.295	23.06	0.09449	0.002078
4.874	24.25	0.12216	0.00311
4.827	20.88	0.13487	0.003201
5.388	31.68	0.09022	0.001738

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
4.936	20.73	0.12181	0.002838
5.385	20.83	0.08798	0.001968
5.736	22.87	0.06618	0.001301
5.32	23.3	0.09497	0.002009
5.048	27.14	0.11165	0.002731
5.075	20.36	0.11297	0.002533
4.833	25.27	0.12103	0.002861
5.751	24.69	0.05839	0.000986
5.495	26.93	0.08464	0.0018
5.304	24.66	0.09163	0.001974
5.473	23.99	0.08109	0.001684
5.545	27.28	0.08428	0.001463
5.226	29.37	0.09869	0.002045
5.249	28.43	0.09227	0.002018
5.379	30.53	0.08008	0.001679
5.751	20.77	0.06505	0.001065
5.715	27.91	0.06382	0.001148
5.339	21.83	0.09104	0.001929
5.354	18.98	0.08577	0.001797
5.612	24.28	0.07364	0.001208
5.21	24.54	0.10282	0.002047
5.666	21.82	0.07605	0.001308
5.053	21.3	0.10796	0.00234
5.335	30.14	0.08451	0.001597
5.7	29.08	0.06876	0.00116
4.699	24.92	0.13044	0.003182
5.352	19.52	0.0859	0.001823
5.634	30.22	0.0669	0.001176
5.238	26.49	0.09657	0.001951
5.426	20.38	0.08348	0.001687
5.746	29.68	0.05998	0.000934
5.015	21.71	0.11631	0.002756

Rows 271–315 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.471	21.29	0.07861	0.001352
5.12	24.06	0.10932	0.002454
5.604	29.69	0.06852	0.001203
5.328	18.54	0.09151	0.00171
5.61	23.53	0.06696	0.000997
4.96	23.74	0.11131	0.002572
5.455	31.76	0.08062	0.001493
5.801	28.22	0.0636	0.001119
5.258	20.6	0.09825	0.002007
5.617	29.24	0.07188	0.001154
5.083	31.21	0.10849	0.002659
5.683	18.49	0.07273	0.001167
5.661	30.84	0.07471	0.000995
5.062	30.14	0.10583	0.002443
5.593	28.93	0.07649	0.001077
5.286	25.38	0.09014	0.00204
5.566	19.65	0.07217	0.001283
5.402	23.53	0.08813	0.001808
5.092	27.49	0.10845	0.002359
5.554	28.11	0.07177	0.001372
5.183	24.64	0.10556	0.002182
5.254	21.98	0.09609	0.002096
5.793	22.22	0.0612	0.000856
5.465	29.61	0.08123	0.001445
5.528	24.94	0.07539	0.001393
5.696	23.55	0.07111	0.001076

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.579	30.13	0.07188	0.001125
5.361	21.44	0.08416	0.001816
5.749	26.9	0.05504	0.001011
5.306	26.97	0.09426	0.001884
5.548	26.1	0.07064	0.00109
5.696	22.65	0.06826	0.000969
4.821	19.0	0.13152	0.003109
4.892	26.12	0.11908	0.002819
5.7	25.31	0.06266	0.001049
5.126	20.38	0.10175	0.002301
5.014	29.79	0.11153	0.002771
5.656	28.71	0.06612	0.001177
5.503	22.13	0.07787	0.001501
5.469	26.54	0.08311	0.001715
5.501	27.56	0.07441	0.001287
5.201	21.11	0.09604	0.002344
5.047	25.64	0.11097	0.002426
5.394	30.99	0.08387	0.001825
5.504	19.01	0.08618	0.001449

Rows 316–360 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.764	24.56	0.06814	0.000918
5.03	19.33	0.11812	0.002776
5.528	18.45	0.0748	0.001247
5.735	26.77	0.06441	0.000989
5.257	21.23	0.09991	0.002109
5.629	31.98	0.06663	0.00123
5.082	26.15	0.11145	0.002562
5.365	30.04	0.08892	0.001847
5.342	22.24	0.09184	0.001926
5.783	29.18	0.06005	0.000862
5.418	25.08	0.08779	0.001715
5.548	22.63	0.07579	0.001334
5.621	21.19	0.07012	0.001057
4.818	20.48	0.13132	0.003174
4.865	26.14	0.12373	0.003088
4.902	24.23	0.11468	0.002886
5.643	18.15	0.07317	0.001273
5.406	20.84	0.08679	0.001711
5.401	24.92	0.08625	0.001757
5.131	18.64	0.10807	0.002235
4.708	27.02	0.1243	0.003139
5.66	19.44	0.07061	0.001113
5.059	21.13	0.10741	0.002588
4.941	30.96	0.11632	0.002689
4.812	31.33	0.12541	0.003082
5.796	31.59	0.06169	0.001004
5.72	26.61	0.06672	0.001205
4.897	19.05	0.1236	0.002682
5.318	22.37	0.09142	0.00186
5.235	21.07	0.10085	0.002156
5.516	30.03	0.07432	0.001169
5.033	30.97	0.1083	0.002365
5.779	19.24	0.05991	0.000915
4.933	28.55	0.12327	0.002816
5.678	29.8	0.07065	0.001017
5.468	30.39	0.07729	0.001434
4.925	18.66	0.12149	0.002836
5.317	25.31	0.09368	0.002009
5.404	27.12	0.07675	0.00158

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.092	23.52	0.10911	0.002522
5.347	18.31	0.09113	0.001763
5.209	28.96	0.09641	0.002041
5.449	21.99	0.08584	0.001619
5.831	18.59	0.0639	0.001096
4.891	31.83	0.11475	0.002624

Rows 361–405 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.692	24.72	0.07485	0.001193
4.934	25.06	0.11752	0.002879
5.744	31.6	0.06289	0.001001
5.623	21.91	0.06147	0.001149
5.431	20.61	0.09168	0.001776
5.26	27.52	0.10065	0.00203
5.127	22.88	0.11559	0.002425
5.158	26.01	0.10218	0.00221
5.208	24.92	0.1061	0.002488
5.584	29.3	0.0724	0.001091
5.504	31.9	0.08041	0.00133
5.356	20.49	0.09335	0.001834
5.734	18.55	0.06178	0.000905
5.104	20.7	0.10984	0.002484
5.136	19.26	0.1064	0.002228
5.632	31.0	0.07252	0.001175
5.165	28.71	0.10517	0.002308
5.282	29.29	0.09623	0.001922
4.944	28.78	0.11546	0.002834
4.796	29.19	0.12934	0.002968
4.895	23.8	0.12425	0.003013
5.111	23.95	0.10943	0.002479
5.742	22.59	0.06715	0.001023
5.082	20.13	0.11816	0.002487
5.297	21.59	0.10526	0.002073
5.745	19.99	0.06393	0.001076
5.695	21.69	0.07085	0.0014
5.381	18.82	0.08547	0.001814
4.854	21.62	0.12848	0.002984
5.759	20.6	0.0659	0.000979
5.525	20.19	0.07556	0.001456
5.355	19.23	0.10196	0.001803
5.734	22.97	0.06043	0.000997
5.014	21.55	0.11783	0.002745
4.798	22.75	0.12875	0.00297
5.691	30.27	0.06132	0.000951
5.227	22.16	0.09888	0.002137
5.087	20.76	0.11096	0.002532
5.194	25.47	0.09823	0.002144
5.476	25.31	0.08328	0.001673
4.82	27.78	0.12685	0.003105
4.845	19.77	0.12517	0.002994
5.501	26.81	0.07374	0.001311
5.622	18.51	0.07171	0.001215
5.693	23.88	0.06545	0.00102

Rows 406–450 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.505	19.31	0.07943	0.001385
5.001	31.61	0.10858	0.002459
5.158	18.55	0.10336	0.002351

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
4.818	31.28	0.1261	0.00316
5.376	22.55	0.09301	0.001925
5.464	31.93	0.07942	0.001581
5.008	22.02	0.11874	0.002709
5.328	18.46	0.08939	0.001959
4.87	31.8	0.11638	0.003045
5.34	26.13	0.08524	0.001754
4.955	27.66	0.11974	0.002964
5.35	20.38	0.09095	0.001815
5.163	18.93	0.11727	0.002399
4.863	18.79	0.12327	0.002803
5.41	29.14	0.09147	0.001707
5.806	29.63	0.06037	0.001055
5.725	27.59	0.06526	0.001108
5.192	24.47	0.10466	0.002293
4.981	20.1	0.10543	0.002664
4.971	29.87	0.11232	0.002791
4.756	28.67	0.13571	0.003196
5.617	26.29	0.07416	0.001084
4.811	31.88	0.12504	0.003027
4.949	26.35	0.1213	0.002835
5.663	21.46	0.06784	0.00112
5.226	27.55	0.09957	0.002241
5.486	19.5	0.09304	0.001768
4.88	19.73	0.12652	0.003066
4.871	28.16	0.12221	0.002836
5.47	29.59	0.08296	0.001383
5.182	31.4	0.09671	0.002306
5.279	31.97	0.0945	0.001825
5.352	26.0	0.08596	0.001734
5.452	27.2	0.08721	0.001659
5.586	19.42	0.07508	0.0012
5.672	19.27	0.0739	0.000991
5.614	26.31	0.07333	0.001231
5.198	28.13	0.09228	0.002089
4.832	20.7	0.13446	0.003065
5.428	19.48	0.09475	0.001797
5.212	29.24	0.0947	0.002139
4.823	30.37	0.12493	0.002914
5.06	24.98	0.11194	0.002571
5.794	21.88	0.06255	0.001065
5.733	26.99	0.06626	0.000969

Rows 451–495 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
4.836	21.26	0.11831	0.002896
5.018	21.01	0.11233	0.002712
5.206	30.77	0.09693	0.002169
5.683	26.84	0.05927	0.001076
5.518	31.41	0.07391	0.001374
5.211	25.14	0.10008	0.002236
5.174	31.58	0.09801	0.002268
5.167	24.99	0.10612	0.00221
5.163	18.35	0.11078	0.002553
4.852	27.5	0.13081	0.003111
5.435	18.58	0.09104	0.001659
5.806	25.81	0.06011	0.001067
5.324	27.75	0.09167	0.001705
5.737	20.72	0.07081	0.001136
5.146	24.32	0.10516	0.002306
5.704	19.74	0.0652	0.001032

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
4.9	31.2	0.1213	0.002911
5.539	28.55	0.07736	0.001403
5.388	27.32	0.07734	0.001553
5.563	28.82	0.06862	0.000981
5.067	29.52	0.11321	0.002603
5.837	31.05	0.05809	0.00099
5.334	22.04	0.09422	0.001776
5.354	26.82	0.09076	0.00178
5.361	29.52	0.0882	0.001985
5.607	30.08	0.07587	0.001219
5.578	27.3	0.07303	0.00146
5.291	28.42	0.0926	0.002128
5.157	19.48	0.09899	0.002284
5.244	21.44	0.09698	0.00207
5.746	26.05	0.06101	0.001137
5.276	24.61	0.09233	0.00199
5.289	28.44	0.08891	0.001841
5.685	26.16	0.068	0.001229
5.028	21.57	0.11269	0.002487
5.22	31.82	0.1004	0.002073
5.259	27.64	0.09926	0.002098
5.602	24.56	0.07092	0.001309
4.988	31.61	0.11012	0.002499
5.649	25.04	0.0666	0.001139
5.396	29.3	0.08209	0.001629
4.916	28.99	0.11506	0.002654
5.71	22.18	0.06925	0.001206
5.119	20.54	0.11126	0.002612
5.35	29.82	0.08753	0.001619

Rows 496–540 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.169	20.89	0.10324	0.002108
4.857	28.14	0.12193	0.002986
5.687	30.18	0.06855	0.001109
5.389	30.87	0.08899	0.001659
4.902	24.23	0.12406	0.003038
5.809	29.19	0.05937	0.001001
5.543	27.08	0.07434	0.001244
4.974	19.45	0.12236	0.002987
5.041	27.62	0.11038	0.002494
5.298	19.05	0.09802	0.002095
5.652	26.77	0.0675	0.00105
5.701	24.89	0.07125	0.001278
4.842	22.6	0.12511	0.002978
4.964	30.23	0.11114	0.00271
4.735	21.96	0.13253	0.003063
5.751	27.51	0.06657	0.000976
5.767	24.51	0.05965	0.000919
5.642	24.32	0.07834	0.001364
5.513	27.1	0.07702	0.001608
4.803	24.06	0.12379	0.003204
5.045	21.26	0.1161	0.00266
5.212	24.04	0.09892	0.002298
5.561	27.25	0.07715	0.001283
5.246	22.45	0.0959	0.002073
4.954	25.8	0.11714	0.002853
5.388	18.28	0.08081	0.00155
5.085	25.41	0.10788	0.002389
4.912	22.61	0.12622	0.002843
4.822	19.52	0.1309	0.003192

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.866	26.54	0.06823	0.000893
5.821	26.72	0.06259	0.000965
5.754	31.96	0.06257	0.00099
5.8	30.92	0.05847	0.000881
4.925	27.43	0.11985	0.002743
5.122	18.48	0.11088	0.002708
4.779	27.44	0.12547	0.003031
5.492	23.11	0.0763	0.001318
5.084	30.35	0.11268	0.002415
5.111	21.42	0.10612	0.002168
5.019	20.36	0.11547	0.002649
5.187	31.41	0.10137	0.002272
5.614	30.79	0.0707	0.001124
4.956	21.47	0.11718	0.002746
5.73	26.29	0.06173	0.000912
5.13	26.03	0.107	0.002264

Rows 541–585 of 600

pH	temperature_C	citric_percent_wv	ascorbic_percent_wv
5.493	31.03	0.07906	0.001356
5.475	28.03	0.07978	0.001284
5.184	26.49	0.11259	0.002331
4.811	20.24	0.1237	0.003036
5.684	25.69	0.07132	0.001053
5.466	31.71	0.08206	0.001523
5.329	28.44	0.08746	0.00157
5.5	19.11	0.08667	0.001649
5.717	27.77	0.06717	0.001129
5.713	22.57	0.06284	0.000821
5.281	22.4	0.09336	0.001952
5.631	23.29	0.0698	0.001067
5.015	31.8	0.10466	0.002585
4.812	26.9	0.12853	0.003115
5.611	28.49	0.06905	0.001152

Rows 586–600 of 600

2. ML_2 — Sugar model

Model type & configuration

Model type	Single-output RandomForestRegressor (scikit-learn)
Saved artifact	ML_2/models/random_forest_sugar_sensors.pkl
Hyperparameters	n_estimators=200, max_depth=12, random_state=42
Input features	pH, TDS, temperature, turbidity
Output target	sugar_pct (Brix)
Training CSV	ML_2/dataset/combined_sensors_sugar.csv (482 rows)
Data origin	400 literature Brix–TDS rows + 82 calibration samples

Full source code — ML_2

Code appears only inside black panels below. Surrounding page, margins, and headings stay white.

ML_2/load_dataset.py

```
ML_2/load_dataset.py

"""
Load combined dataset: pH, TDS, temperature, turbidity -> sugar_pct.
"""
from pathlib import Path
from typing import Tuple

import numpy as np
import pandas as pd

FEATURE_COLUMNS = ["pH", "TDS", "temperature", "turbidity"]
TARGET_COLUMN = "sugar_pct"

def get_dataset_path() -> Path:
    return Path(__file__).resolve().parent / "dataset" / "combined_sensors_sugar.csv"

def load_dataset(csv_path: str | Path | None = None) -> pd.DataFrame:
    path = Path(csv_path) if csv_path else get_dataset_path()
    if not path.exists():
        raise FileNotFoundError(
            f"Dataset not found: {path}. Run: python ML_2/merge_datasets.py"
        )
    df = pd.read_csv(path)
    df.columns = df.columns.str.strip()
    return df

def prepare_xy(df: pd.DataFrame) -> Tuple[np.ndarray, np.ndarray]:
    for col in FEATURE_COLUMNS + [TARGET_COLUMN]:
        if col not in df.columns:
            raise ValueError(f"Missing column: {col}")
    df_clean = df[FEATURE_COLUMNS + [TARGET_COLUMN]].dropna()
    X = df_clean[FEATURE_COLUMNS].astype(np.float64).values
    y = df_clean[TARGET_COLUMN].astype(np.float64).values
    return X, y

def load_xy(csv_path: str | Path | None = None) -> Tuple[np.ndarray, np.ndarray]:
    df = load_dataset(csv_path)
    return prepare_xy(df)
```

ML_2/merge_datasets.py

```
ML_2/merge_datasets.py

"""
Merge literature Brix?TDS synthetic data + ml calibration CSV into one training file.

Sources:
- ML_2/dataset/literature_tds_sugar_synthetic.csv (from generate_sugar_tds_literature_dataset.py)
- ml/data/calibration/calibration_dataset.csv      (existing natural + artificial samples)

Output:
- ML_2/dataset/combined_sensors_sugar.csv

Run from project root: python ML_2/merge_datasets.py
"""
from pathlib import Path

import pandas as pd

PROJECT_ROOT = Path(__file__).resolve().parent.parent
LITERATURE_PATH = Path(__file__).resolve().parent / "dataset" / "literature_tds_sugar_synthetic.csv"
CALIBRATION_PATH = PROJECT_ROOT / "ml" / "data" / "calibration" / "calibration_dataset.csv"
OUT_PATH = Path(__file__).resolve().parent / "dataset" / "combined_sensors_sugar.csv"

CANONICAL_COLUMNS = [
    "sample_id",
    "pH",
    "TDS",
    "temperature",
    "turbidity",
    "sugar_pct",
    "citric_acid_pct",
    "ascorbic_acid_pct",
    "source",
    "notes",
]

def _normalize(df: pd.DataFrame, source: str) -> pd.DataFrame:
    df = df.copy()
    df.columns = df.columns.str.strip()
    rename = {}
    if "temperature_C" in df.columns and "temperature" not in df.columns:
        rename["temperature_C"] = "temperature"
    df = df.rename(columns=rename)

    for col in ["pH", "TDS", "temperature", "turbidity", "sugar_pct"]:
        if col not in df.columns:
            raise ValueError(f"{source}: missing column {col}")

    if "citric_acid_pct" not in df.columns:
        df["citric_acid_pct"] = 0.0
    if "ascorbic_acid_pct" not in df.columns:
        df["ascorbic_acid_pct"] = 0.0
    if "sample_id" not in df.columns:
        df["sample_id"] = [f"{source[:3].upper()}{i:04d}" for i in range(len(df))]
    if "notes" not in df.columns:
        df["notes"] = ""
```

ML_2/merge_datasets.py (cont.)

```
    df["source"] = source
    return df[CANONICAL_COLUMNS]

def merge_all() -> pd.DataFrame:
    frames: list[pd.DataFrame] = []

    if LITERATURE_PATH.exists():
        lit = pd.read_csv(LITERATURE_PATH)
        frames.append(_normalize(lit, "literature"))
    else:
        print(f"Warning: {LITERATURE_PATH} not found. Run generate_sugar_tds_literature_dataset.py first.")

    if CALIBRATION_PATH.exists():
        cal = pd.read_csv(CALIBRATION_PATH)
        frames.append(_normalize(cal, "calibration"))
    else:
        print(f"Warning: {CALIBRATION_PATH} not found.")

    if not frames:
        raise FileNotFoundError("No input datasets found.")

    combined = pd.concat(frames, ignore_index=True)
    combined = combined.drop_duplicates(
        subset=["pH", "TDS", "temperature", "turbidity", "sugar_pct"],
        keep="first",
    )

    OUT_PATH.parent.mkdir(parents=True, exist_ok=True)
    combined.to_csv(OUT_PATH, index=False)
    print(f"Saved {len(combined)} rows to {OUT_PATH}")
    print(f"  literature: {(combined['source'] == 'literature').sum()}")
    print(f"  calibration: {(combined['source'] == 'calibration').sum()}")
    print(combined[["pH", "TDS", "sugar_pct", "source"]].head())
    return combined

if __name__ == "__main__":
    merge_all()
```

ML_2/train_model.py

```

ML_2/train_model.py

"""
Train Random Forest: pH, TDS, temperature, turbidity -> sugar_pct.

Run from project root:
python ML_2/merge_datasets.py
python ML_2/train_model.py
"""

from pathlib import Path

import joblib
import numpy as np
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.model_selection import train_test_split

try:
    from load_dataset import TARGET_COLUMN, load_xy
except ImportError:
    from ML_2.load_dataset import TARGET_COLUMN, load_xy

RANDOM_STATE = 42
TEST_SIZE = 0.2
MODEL_PARAMS = {"n_estimators": 200, "max_depth": 12, "random_state": RANDOM_STATE}

def get_models_dir() -> Path:
    return Path(__file__).resolve().parent / "models"

def train_and_evaluate(csv_path: str | Path | None = None) -> RandomForestRegressor:
    X, y = load_xy(csv_path)
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=TEST_SIZE, random_state=RANDOM_STATE
    )
    model = RandomForestRegressor(**MODEL_PARAMS)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    r2 = r2_score(y_test, y_pred)
    rmse = np.sqrt(mean_squared_error(y_test, y_pred))
    mae = mean_absolute_error(y_test, y_pred)

    print("Evaluation (test set) ? sugar prediction:")
    print("-" * 50)
    print(f"    {TARGET_COLUMN}: R2={r2:.4f}, RMSE={rmse:.4f}, MAE={mae:.4f}")
    print("-" * 50)
    return model

def save_model(model: RandomForestRegressor, path: str | Path | None = None) -> Path:
    out_dir = get_models_dir()
    out_dir.mkdir(parents=True, exist_ok=True)
    out_path = path or out_dir / "random_forest_sugar_sensors.pkl"
    joblib.dump(model, out_path)
    print(f"Model saved to {out_path}")

```

```

ML_2/train_model.py (cont.)

    return Path(out_path)

if __name__ == "__main__":
    model = train_and_evaluate()
    save_model(model)

```

```

ML_2/predict.py

"""
Prediction: pH, TDS, temperature, turbidity -> sugar_pct.
"""

from pathlib import Path
from typing import Any

import joblib
import numpy as np

_MODEL: Any = None
_MODEL_PATH = Path(__file__).resolve().parent / "models" / "random_forest_sugar_sensors.pkl"

def _get_model() -> Any:
    global _MODEL
    if _MODEL is None:
        if not _MODEL_PATH.exists():
            raise FileNotFoundError(
                f"Model not found: {_MODEL_PATH}. Run: python ML_2/train_model.py"
            )
        _MODEL = joblib.load(_MODEL_PATH)
    return _MODEL

def predict(
    pH: float,
    tds: float,
    temperature: float,
    turbidity: float,
) -> dict[str, float]:
    """Predict sugar_pct from four sensor readings."""
    model = _get_model()
    X = np.array([[pH, tds, temperature, turbidity]], dtype=np.float64)
    sugar = float(model.predict(X)[0])
    return {
        "pH": float(pH),
        "TDS": float(tds),
        "temperature": float(temperature),
        "turbidity": float(turbidity),
        "sugar_pct": sugar,
    }

```

ML_2 full training dataset

File: **ML_2/dataset/combined_sensors_sugar.csv** — all training rows (white table surface).

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0001	5.71	4340.7	17.3	13.0	13.1766	0.0705	0.000925	literature	Literature_Brix_TDS_t18.6h
LIT0002	5.86	3267.1	25.0	17.5	5.0858	0.0638	0.000881	literature	Literature_Brix_TDS_t3.1h
LIT0003	5.71	3569.2	26.6	13.7	8.9919	0.0752	0.000941	literature	Literature_Brix_TDS_t13.3h
LIT0004	5.68	4333.1	18.2	20.0	13.5634	0.0805	0.000979	literature	Literature_Brix_TDS_t18.7h
LIT0005	5.84	3176.2	17.8	11.2	5.6927	0.0601	0.001002	literature	Literature_Brix_TDS_t7.8h
LIT0006	5.75	3144.7	21.4	16.4	8.0588	0.0713	0.000968	literature	Literature_Brix_TDS_t10.5h
LIT0007	5.84	3174.7	25.9	12.3	5.6234	0.0614	0.000899	literature	Literature_Brix_TDS_t4.8h
LIT0008	5.87	3054.2	23.9	11.8	5.9426	0.0621	0.000997	literature	Literature_Brix_TDS_t3.4h
LIT0009	5.68	3435.8	19.0	15.7	9.6329	0.0666	0.000911	literature	Literature_Brix_TDS_t13.4h
LIT0010	5.88	3199.6	23.8	13.1	5.3489	0.0637	0.000925	literature	Literature_Brix_TDS_t1.4h
LIT0011	5.62	4375.3	26.1	20.5	14.631	0.0729	0.001113	literature	Literature_Brix_TDS_t19.5h
LIT0012	5.81	3234.2	20.2	12.9	5.6132	0.0583	0.000834	literature	Literature_Brix_TDS_t3.7h
LIT0013	5.77	3212.6	19.7	18.1	5.005	0.0657	0.001009	literature	Literature_Brix_TDS_t2.8h
LIT0014	5.72	3125.2	22.4	8.7	8.0166	0.0715	0.000969	literature	Literature_Brix_TDS_t10.8h
LIT0015	5.77	3155.6	22.0	13.5	5.0242	0.0732	0.000931	literature	Literature_Brix_TDS_t4.2h
LIT0016	5.91	3201.1	20.6	14.1	5.3161	0.0659	0.000855	literature	Literature_Brix_TDS_t2.0h
LIT0017	5.72	3102.0	24.3	12.5	5.1875	0.0635	0.000869	literature	Literature_Brix_TDS_t4.1h
LIT0018	5.83	4527.1	29.1	14.3	14.1236	0.0553	0.000971	literature	Literature_Brix_TDS_t18.8h
LIT0019	5.76	3027.1	27.6	15.3	5.8493	0.0669	0.000867	literature	Literature_Brix_TDS_t8.0h
LIT0020	5.93	3161.0	26.8	16.8	5.1555	0.058	0.000855	literature	Literature_Brix_TDS_t2.6h
LIT0021	5.79	3444.2	27.3	8.7	4.8584	0.0695	0.000981	literature	Literature_Brix_TDS_t0.9h
LIT0022	5.75	3266.6	16.6	11.2	9.0761	0.063	0.000964	literature	Literature_Brix_TDS_t12.4h
LIT0023	5.68	4129.9	20.4	18.9	12.7601	0.0701	0.001095	literature	Literature_Brix_TDS_t18.0h
LIT0024	5.75	3193.6	19.4	19.4	5.8209	0.0748	0.000936	literature	Literature_Brix_TDS_t2.3h
LIT0025	5.75	3013.4	26.9	11.9	6.0493	0.0679	0.000937	literature	Literature_Brix_TDS_t4.3h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0026	5.89	3206.7	17.7	11.9	5.5344	0.0574	0.000852	literature	Literature_Brix_TDS_t0.3h
LIT0027	5.72	3250.0	17.8	11.7	5.4824	0.0683	0.00103	literature	Literature_Brix_TDS_t4.8h
LIT0028	5.64	3770.1	16.3	18.3	10.4453	0.0745	0.00099	literature	Literature_Brix_TDS_t15.9h
LIT0029	5.93	3460.0	29.1	12.0	5.5009	0.0541	0.000903	literature	Literature_Brix_TDS_t3.4h
LIT0030	5.81	3107.2	19.3	11.5	5.4756	0.0608	0.000908	literature	Literature_Brix_TDS_t0.4h
LIT0031	5.65	3137.7	22.4	23.5	7.1287	0.0682	0.000973	literature	Literature_Brix_TDS_t9.0h
LIT0032	5.62	4289.0	18.0	25.4	15.9635	0.0709	0.001092	literature	Literature_Brix_TDS_t19.6h
LIT0033	5.61	5138.4	17.6	18.4	16.9667	0.0744	0.001153	literature	Literature_Brix_TDS_t22.1h
LIT0034	5.85	3305.5	20.5	12.6	5.3252	0.0629	0.000892	literature	Literature_Brix_TDS_t1.3h
LIT0035	5.93	3438.1	21.8	9.3	5.0964	0.0556	0.001045	literature	Literature_Brix_TDS_t1.0h
LIT0036	5.89	2975.1	28.6	8.0	5.3239	0.0585	0.000971	literature	Literature_Brix_TDS_t1.2h
LIT0037	5.67	3715.0	23.2	23.0	9.7526	0.0618	0.000998	literature	Literature_Brix_TDS_t15.4h
LIT0038	5.58	3674.1	21.8	24.0	9.941	0.0708	0.001254	literature	Literature_Brix_TDS_t15.3h
LIT0039	5.75	2975.4	29.3	17.0	5.961	0.0751	0.001044	literature	Literature_Brix_TDS_t7.9h
LIT0040	5.77	3459.0	23.5	16.5	9.823	0.0644	0.001029	literature	Literature_Brix_TDS_t12.9h
LIT0041	5.87	3116.4	17.3	20.9	5.881	0.0645	0.000824	literature	Literature_Brix_TDS_t4.9h
LIT0042	5.91	3096.6	27.0	10.3	5.002	0.0559	0.000953	literature	Literature_Brix_TDS_t3.3h
LIT0043	5.77	3607.7	24.0	8.0	9.3056	0.0681	0.000956	literature	Literature_Brix_TDS_t14.2h
LIT0044	5.61	4109.2	17.7	12.1	14.9407	0.0812	0.001158	literature	Literature_Brix_TDS_t19.2h
LIT0045	5.81	3300.0	28.6	16.3	7.3351	0.0659	0.000881	literature	Literature_Brix_TDS_t10.0h

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0046	5.83	3151.7	22.1	14.0	5.8899	0.0696	0.000891	literature	Literature_Brix_TDS_t7.5h
LIT0047	5.62	5403.2	29.8	22.9	17.1129	0.0786	0.001169	literature	Literature_Brix_TDS_t22.5h
LIT0048	5.73	4393.7	25.6	19.2	15.6862	0.0742	0.000822	literature	Literature_Brix_TDS_t20.4h
LIT0049	5.8	3016.1	28.9	15.5	5.3256	0.0596	0.000765	literature	Literature_Brix_TDS_t7.1h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0050	5.67	4912.9	30.0	16.4	16.0229	0.0728	0.001081	literature	Literature_Brix_TDS_t2 1.1h
LIT0051	5.61	4912.3	26.9	22.5	16.641	0.0737	0.00117	literature	Literature_Brix_TDS_t2 1.5h
LIT0052	5.72	3288.7	27.2	8.0	5.0673	0.0602	0.000896	literature	Literature_Brix_TDS_t0 .9h
LIT0053	5.72	4387.8	27.7	18.1	14.3321	0.0753	0.000957	literature	Literature_Brix_TDS_t1 9.2h
LIT0054	5.58	4461.2	19.8	22.0	15.5589	0.0839	0.001076	literature	Literature_Brix_TDS_t2 0.4h
LIT0055	5.83	3240.6	25.9	16.1	4.7993	0.0614	0.000907	literature	Literature_Brix_TDS_t1 .3h
LIT0056	5.59	5695.6	24.0	21.3	17.4714	0.0817	0.001141	literature	Literature_Brix_TDS_t2 3.5h
LIT0057	5.77	3494.3	18.1	16.1	9.8416	0.0682	0.001106	literature	Literature_Brix_TDS_t1 3.9h
LIT0058	5.88	3183.5	22.0	14.3	5.3819	0.0607	0.000975	literature	Literature_Brix_TDS_t2 .9h
LIT0059	5.68	2941.8	16.5	14.1	6.8017	0.0795	0.001093	literature	Literature_Brix_TDS_t9 .8h
LIT0060	5.69	3468.5	25.1	19.1	9.5564	0.0699	0.001073	literature	Literature_Brix_TDS_t1 3.7h
LIT0061	5.76	3529.4	26.3	15.8	7.7677	0.0701	0.000945	literature	Literature_Brix_TDS_t1 0.4h
LIT0062	5.67	3412.5	22.9	18.6	8.8488	0.075	0.001114	literature	Literature_Brix_TDS_t1 2.5h
LIT0063	5.82	3237.8	20.4	8.0	5.0665	0.0631	0.000791	literature	Literature_Brix_TDS_t1 .3h
LIT0064	5.59	4788.5	17.7	8.0	16.3256	0.0743	0.001364	literature	Literature_Brix_TDS_t2 0.6h
LIT0065	5.56	4298.7	21.4	18.6	13.2836	0.0778	0.001288	literature	Literature_Brix_TDS_t1 8.5h
LIT0066	5.77	3292.7	29.8	12.3	9.3844	0.0633	0.001057	literature	Literature_Brix_TDS_t1 2.6h
LIT0067	5.9	3104.4	26.6	14.2	5.9161	0.0663	0.000921	literature	Literature_Brix_TDS_t3 .4h
LIT0068	5.71	3200.0	20.1	11.3	5.3858	0.0747	0.000758	literature	Literature_Brix_TDS_t4 .3h
LIT0069	5.84	3075.6	25.1	11.7	4.6621	0.0624	0.000966	literature	Literature_Brix_TDS_t2 .0h
LIT0070	5.78	3219.4	22.8	16.1	6.8463	0.0745	0.000943	literature	Literature_Brix_TDS_t8 .8h
LIT0071	5.55	4970.6	23.0	17.2	16.3795	0.0754	0.001251	literature	Literature_Brix_TDS_t2 1.8h
LIT0072	5.67	5063.4	18.7	18.7	16.778	0.0695	0.00106	literature	Literature_Brix_TDS_t2 2.0h
LIT0073	5.79	3099.6	25.8	15.2	5.4158	0.0601	0.001057	literature	Literature_Brix_TDS_t7 .4h
LIT0074	5.59	3316.6	29.5	14.6	8.5067	0.0726	0.00124	literature	Literature_Brix_TDS_t1 1.5h
LIT0075	5.68	3975.6	29.6	13.0	11.7578	0.0727	0.00115	literature	Literature_Brix_TDS_t1 7.6h
LIT0076	5.59	3951.7	26.9	15.9	12.6011	0.0802	0.001203	literature	Literature_Brix_TDS_t1 8.0h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0077	5.68	5558.4	23.8	17.3	17.8491	0.0738	0.001064	literature	Literature_Brix_TDS_t2 3.5h
LIT0078	5.78	3253.5	25.5	8.0	4.521	0.0683	0.000885	literature	Literature_Brix_TDS_t0 .9h
LIT0079	5.63	4134.6	24.7	16.2	12.6511	0.0747	0.001125	literature	Literature_Brix_TDS_t1 7.7h
LIT0080	5.66	3780.6	25.4	13.3	10.5154	0.0634	0.00118	literature	Literature_Brix_TDS_t1 6.3h
LIT0081	5.62	4423.7	19.8	18.1	15.6583	0.0707	0.001169	literature	Literature_Brix_TDS_t2 0.5h
LIT0082	5.89	3194.2	20.1	13.4	5.6316	0.0601	0.000999	literature	Literature_Brix_TDS_t8 .2h
LIT0083	5.61	5573.4	28.8	27.4	17.5189	0.0748	0.00127	literature	Literature_Brix_TDS_t2 3.9h
LIT0084	5.78	3218.2	24.9	17.4	5.407	0.0651	0.000815	literature	Literature_Brix_TDS_t1 .6h
LIT0085	5.66	4117.2	19.3	11.6	12.4989	0.0749	0.001007	literature	Literature_Brix_TDS_t1 8.0h
LIT0086	5.62	4150.4	19.4	21.3	12.7712	0.0763	0.001306	literature	Literature_Brix_TDS_t1 7.8h
LIT0087	5.76	3464.4	22.7	21.5	9.5046	0.0655	0.000915	literature	Literature_Brix_TDS_t1 3.5h
LIT0088	5.78	3324.3	29.4	14.7	5.8323	0.0657	0.001026	literature	Literature_Brix_TDS_t4 .4h
LIT0089	5.47	5392.4	16.9	16.0	17.1073	0.0825	0.001577	literature	Literature_Brix_TDS_t2 3.0h
LIT0090	5.9	3177.9	25.8	8.8	4.7921	0.0622	0.000932	literature	Literature_Brix_TDS_t1 .2h

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0091	5.68	3284.1	16.0	14.5	7.4942	0.0667	0.000874	literature	Literature_Brix_TDS_t1 0.0h
LIT0092	5.74	3094.9	20.5	8.0	7.6411	0.0628	0.000876	literature	Literature_Brix_TDS_t1 0.4h
LIT0093	5.72	3223.6	22.5	15.6	5.7837	0.0696	0.000994	literature	Literature_Brix_TDS_t5 .5h
LIT0094	5.86	3177.5	16.4	8.0	5.512	0.053	0.00103	literature	Literature_Brix_TDS_t3 .3h
LIT0095	5.71	3284.5	27.9	23.7	9.0581	0.072	0.00092	literature	Literature_Brix_TDS_t1 3.7h
LIT0096	5.58	4636.9	24.4	26.0	15.8364	0.082	0.001129	literature	Literature_Brix_TDS_t2 0.5h
LIT0097	5.85	3157.6	21.8	17.5	6.1791	0.0655	0.000977	literature	Literature_Brix_TDS_t7 .6h
LIT0098	5.69	3049.0	25.9	21.7	6.2659	0.0769	0.001014	literature	Literature_Brix_TDS_t8 .3h
LIT0099	5.64	3605.8	18.4	17.0	9.5672	0.0689	0.001085	literature	Literature_Brix_TDS_t1 4.3h
LIT0100	5.85	3302.3	28.6	12.3	5.6789	0.0621	0.000863	literature	Literature_Brix_TDS_t0 .4h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0101	5.81	2968.0	25.3	8.0	5.5584	0.0642	0.00103	literature	Literature_Brix_TDS_t6.6h
LIT0102	5.74	3135.3	23.7	15.2	6.0529	0.07	0.001053	literature	Literature_Brix_TDS_t5.6h
LIT0103	5.69	3351.5	23.2	15.7	9.3553	0.069	0.001009	literature	Literature_Brix_TDS_t13.6h
LIT0104	5.72	3226.9	25.6	16.4	7.1235	0.0649	0.000912	literature	Literature_Brix_TDS_t9.8h
LIT0105	5.69	2886.1	28.3	13.7	5.1919	0.0709	0.001088	literature	Literature_Brix_TDS_t4.8h
LIT0106	5.6	4258.1	28.7	14.8	12.2661	0.0726	0.001131	literature	Literature_Brix_TDS_t17.5h
LIT0107	5.91	3098.4	21.3	20.6	6.2563	0.0596	0.00096	literature	Literature_Brix_TDS_t4.9h
LIT0108	5.73	3206.4	24.4	13.1	7.07	0.0673	0.001023	literature	Literature_Brix_TDS_t7.7h
LIT0109	5.78	3154.3	25.6	8.0	6.8872	0.0693	0.000871	literature	Literature_Brix_TDS_t9.1h
LIT0110	5.78	3180.4	18.8	8.0	5.2763	0.0646	0.000903	literature	Literature_Brix_TDS_t2.4h
LIT0111	5.83	3001.2	25.1	8.0	5.8759	0.056	0.000949	literature	Literature_Brix_TDS_t6.2h
LIT0112	5.79	3108.2	29.8	9.9	5.8481	0.0612	0.000863	literature	Literature_Brix_TDS_t7.1h
LIT0113	5.67	3106.9	27.8	9.8	5.8491	0.0703	0.001001	literature	Literature_Brix_TDS_t3.3h
LIT0114	5.75	3249.9	26.3	15.9	5.6507	0.0621	0.000981	literature	Literature_Brix_TDS_t6.1h
LIT0115	5.62	4382.7	17.7	17.2	13.0289	0.0697	0.001119	literature	Literature_Brix_TDS_t18.2h
LIT0116	5.67	4650.2	19.4	22.7	15.4341	0.0769	0.001137	literature	Literature_Brix_TDS_t20.3h
LIT0117	5.7	3166.9	29.2	16.7	9.6623	0.0685	0.000959	literature	Literature_Brix_TDS_t12.7h
LIT0118	5.71	3338.4	17.8	15.9	8.7715	0.0718	0.001043	literature	Literature_Brix_TDS_t11.8h
LIT0119	5.54	5505.7	19.5	23.4	17.4064	0.0828	0.001289	literature	Literature_Brix_TDS_t22.9h
LIT0120	5.79	3247.6	23.4	15.9	6.1364	0.0543	0.000981	literature	Literature_Brix_TDS_t7.7h
LIT0121	5.69	4169.0	20.8	15.1	11.8701	0.0766	0.000943	literature	Literature_Brix_TDS_t17.8h
LIT0122	5.79	3717.1	28.9	13.0	9.7356	0.0654	0.000947	literature	Literature_Brix_TDS_t14.9h
LIT0123	5.66	5659.2	17.8	18.8	17.811	0.0726	0.001067	literature	Literature_Brix_TDS_t23.8h
LIT0124	5.69	5065.5	25.2	13.7	16.4779	0.0732	0.001032	literature	Literature_Brix_TDS_t22.1h
LIT0125	5.79	3199.5	16.6	9.1	6.2492	0.0674	0.000926	literature	Literature_Brix_TDS_t8.2h
LIT0126	5.79	3957.5	26.3	16.8	10.2647	0.0672	0.001011	literature	Literature_Brix_TDS_t15.8h
LIT0127	5.72	3425.7	20.8	14.4	9.0994	0.0722	0.00095	literature	Literature_Brix_TDS_t12.2h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0128	5.87	3127.5	16.3	8.0	5.5097	0.0672	0.00091	literature	Literature_Brix_TDS_t2.2h
LIT0129	5.64	4590.4	19.9	17.5	15.5437	0.0765	0.001112	literature	Literature_Brix_TDS_t20.7h
LIT0130	5.5	5174.5	29.3	10.6	16.7983	0.082	0.001412	literature	Literature_Brix_TDS_t21.8h
LIT0131	5.74	3128.2	16.3	12.5	6.3771	0.0686	0.001029	literature	Literature_Brix_TDS_t7.7h
LIT0132	5.68	3314.0	23.6	18.3	8.273	0.0721	0.000942	literature	Literature_Brix_TDS_t10.7h
LIT0133	5.73	3191.8	29.7	9.0	5.5444	0.0682	0.000959	literature	Literature_Brix_TDS_t5.9h
LIT0134	5.74	3453.4	20.1	16.3	8.1689	0.0704	0.000938	literature	Literature_Brix_TDS_t11.5h
LIT0135	5.76	3270.2	16.1	9.5	5.4968	0.0695	0.001066	literature	Literature_Brix_TDS_t5.3h

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0136	5.8	3024.7	30.0	9.5	5.3461	0.0662	0.001056	literature	Literature_Brix_TDS_t3.3h
LIT0137	5.71	3831.2	17.9	18.7	9.5127	0.0704	0.000959	literature	Literature_Brix_TDS_t14.9h
LIT0138	5.72	3605.1	21.9	14.9	9.4914	0.0695	0.001014	literature	Literature_Brix_TDS_t15.9h
LIT0139	5.78	3905.2	26.2	21.1	9.6865	0.0713	0.00104	literature	Literature_Brix_TDS_t15.7h
LIT0140	5.68	3272.1	27.0	10.0	7.8554	0.0687	0.001114	literature	Literature_Brix_TDS_t10.2h
LIT0141	5.7	3070.9	28.5	15.0	6.4852	0.071	0.001059	literature	Literature_Brix_TDS_t7.7h
LIT0142	5.84	3035.3	16.0	10.2	5.5838	0.0623	0.000935	literature	Literature_Brix_TDS_t3.7h
LIT0143	5.67	3745.0	28.1	13.6	10.5335	0.0711	0.000966	literature	Literature_Brix_TDS_t15.7h
LIT0144	5.83	3326.8	26.5	12.4	5.2205	0.0637	0.001088	literature	Literature_Brix_TDS_t2.2h
LIT0145	5.76	3076.3	17.7	16.8	5.9217	0.066	0.001032	literature	Literature_Brix_TDS_t8.8h
LIT0146	5.66	4572.6	25.1	21.4	15.227	0.0733	0.001043	literature	Literature_Brix_TDS_t19.8h
LIT0147	5.53	5439.5	25.2	18.8	16.7761	0.0737	0.001217	literature	Literature_Brix_TDS_t23.4h
LIT0148	5.69	3668.9	16.7	11.8	10.1656	0.0637	0.000983	literature	Literature_Brix_TDS_t13.2h
LIT0149	5.7	3010.4	21.4	16.8	7.5061	0.0724	0.000864	literature	Literature_Brix_TDS_t10.0h
LIT0150	5.71	3568.2	19.0	15.2	9.1045	0.0717	0.001002	literature	Literature_Brix_TDS_t13.5h
LIT0151	5.61	5065.0	16.8	26.9	16.8901	0.0833	0.001035	literature	Literature_Brix_TDS_t21.4h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0152	5.6	3727.0	21.9	19.1	10.9441	0.0757	0.001103	literature	Literature_Brix_TDS_t16.3h
LIT0153	5.91	2998.3	19.7	18.1	5.7119	0.058	0.000935	literature	Literature_Brix_TDS_t7.7h
LIT0154	5.51	5086.0	24.7	14.7	16.0169	0.0865	0.00137	literature	Literature_Brix_TDS_t22.3h
LIT0155	5.59	4537.1	21.6	21.5	15.1975	0.0743	0.001218	literature	Literature_Brix_TDS_t19.7h
LIT0156	5.58	4251.3	22.9	9.3	14.412	0.0777	0.001225	literature	Literature_Brix_TDS_t19.3h
LIT0157	5.74	3622.7	17.4	21.8	9.5178	0.0645	0.001085	literature	Literature_Brix_TDS_t14.0h
LIT0158	5.72	3198.6	29.1	14.6	7.7218	0.0632	0.001027	literature	Literature_Brix_TDS_t9.8h
LIT0159	5.66	3540.0	20.5	19.6	9.1287	0.0725	0.001174	literature	Literature_Brix_TDS_t13.6h
LIT0160	5.86	3161.9	23.9	13.1	5.848	0.0619	0.000883	literature	Literature_Brix_TDS_t4.0h
LIT0161	5.71	5305.8	29.2	17.9	17.1402	0.0717	0.0009	literature	Literature_Brix_TDS_t22.3h
LIT0162	5.87	3393.0	16.4	16.8	4.8039	0.0529	0.000789	literature	Literature_Brix_TDS_t0.1h
LIT0163	5.65	3289.0	28.3	23.7	8.4048	0.0765	0.001069	literature	Literature_Brix_TDS_t11.2h
LIT0164	5.83	2896.3	29.5	8.0	5.3437	0.0614	0.000857	literature	Literature_Brix_TDS_t3.9h
LIT0165	5.74	3332.7	22.7	15.7	9.5844	0.0587	0.00087	literature	Literature_Brix_TDS_t14.3h
LIT0166	5.67	3168.3	27.1	12.7	6.6773	0.0738	0.001065	literature	Literature_Brix_TDS_t10.2h
LIT0167	5.54	4864.9	25.7	16.0	16.044	0.0771	0.001271	literature	Literature_Brix_TDS_t21.0h
LIT0168	5.71	3772.2	19.0	16.5	9.7924	0.073	0.0008	literature	Literature_Brix_TDS_t14.8h
LIT0169	5.85	3203.4	29.6	14.6	5.5972	0.0653	0.001167	literature	Literature_Brix_TDS_t2.3h
LIT0170	5.72	3754.0	23.5	12.9	9.5261	0.079	0.000976	literature	Literature_Brix_TDS_t15.2h
LIT0171	5.66	3951.2	29.1	12.5	11.6119	0.0744	0.0012	literature	Literature_Brix_TDS_t16.9h
LIT0172	5.79	3092.4	21.6	11.7	5.3356	0.0697	0.000811	literature	Literature_Brix_TDS_t3.7h
LIT0173	5.7	3152.0	17.3	19.5	5.7991	0.0743	0.000839	literature	Literature_Brix_TDS_t7.5h
LIT0174	5.6	5419.4	27.1	22.0	17.7802	0.0766	0.001171	literature	Literature_Brix_TDS_t23.4h
LIT0175	5.59	5236.0	21.4	12.8	18.0062	0.0772	0.001279	literature	Literature_Brix_TDS_t23.3h
LIT0176	5.79	3066.9	24.6	10.4	5.35	0.0677	0.0008	literature	Literature_Brix_TDS_t1.9h
LIT0177	5.64	5588.8	29.7	24.5	17.847	0.0726	0.001064	literature	Literature_Brix_TDS_t23.4h
LIT0178	5.85	3264.4	23.5	12.5	4.9868	0.0648	0.000916	literature	Literature_Brix_TDS_t1.5h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0179	5.78	3804.7	17.3	25.9	9.158	0.0705	0.001028	literature	Literature_Brix_TDS_t14.8h
LIT0180	5.78	3128.8	20.5	15.8	6.7335	0.0628	0.001016	literature	Literature_Brix_TDS_t9.1h

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0181	5.78	3220.9	18.6	10.4	5.9151	0.0601	0.000969	literature	Literature_Brix_TDS_t5.5h
LIT0182	5.65	2969.9	22.7	13.4	8.8952	0.0749	0.000955	literature	Literature_Brix_TDS_t12.2h
LIT0183	5.93	3166.4	22.6	12.6	5.6335	0.0576	0.000924	literature	Literature_Brix_TDS_t5.5h
LIT0184	5.73	4405.9	24.9	27.8	15.0962	0.0693	0.001093	literature	Literature_Brix_TDS_t19.7h
LIT0185	5.63	3700.6	22.0	21.3	10.3122	0.0745	0.000991	literature	Literature_Brix_TDS_t15.9h
LIT0186	5.74	3039.5	18.8	15.6	7.7753	0.0594	0.000935	literature	Literature_Brix_TDS_t11.0h
LIT0187	5.76	2943.4	18.8	16.0	6.5514	0.0698	0.000959	literature	Literature_Brix_TDS_t6.5h
LIT0188	5.71	4097.2	22.7	19.4	13.3702	0.0708	0.000901	literature	Literature_Brix_TDS_t18.3h
LIT0189	5.8	3033.2	22.7	14.3	5.1639	0.0609	0.001007	literature	Literature_Brix_TDS_t3.1h
LIT0190	5.82	3162.9	18.5	14.7	7.8701	0.0664	0.000781	literature	Literature_Brix_TDS_t10.8h
LIT0191	5.67	4173.4	22.1	21.6	13.5079	0.0645	0.000898	literature	Literature_Brix_TDS_t18.0h
LIT0192	5.64	4362.7	28.1	13.0	12.9339	0.0629	0.001062	literature	Literature_Brix_TDS_t18.3h
LIT0193	5.69	5216.6	16.2	10.7	16.4806	0.0735	0.001023	literature	Literature_Brix_TDS_t21.8h
LIT0194	5.83	3364.7	19.2	19.5	9.7525	0.0571	0.000876	literature	Literature_Brix_TDS_t13.6h
LIT0195	5.65	4909.5	19.8	24.4	16.0024	0.0727	0.000928	literature	Literature_Brix_TDS_t20.9h
LIT0196	5.62	3518.5	16.8	14.1	9.5349	0.0737	0.001015	literature	Literature_Brix_TDS_t13.4h
LIT0197	5.73	3954.2	19.7	15.9	10.2289	0.0698	0.000885	literature	Literature_Brix_TDS_t16.4h
LIT0198	5.58	4344.6	19.7	21.4	14.5486	0.0711	0.001202	literature	Literature_Brix_TDS_t19.5h
LIT0199	5.57	4256.8	19.0	18.3	13.7206	0.082	0.001038	literature	Literature_Brix_TDS_t18.6h
LIT0200	5.87	3170.0	17.2	11.8	5.5323	0.0549	0.000762	literature	Literature_Brix_TDS_t0.9h
LIT0201	5.72	3088.7	28.1	12.8	5.2218	0.0667	0.00098	literature	Literature_Brix_TDS_t4.5h
LIT0202	5.84	3222.8	26.2	15.1	7.5255	0.0677	0.001051	literature	Literature_Brix_TDS_t9.9h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0203	5.83	3302.8	27.1	12.7	5.4924	0.0622	0.000973	literature	Literature_Brix_TDS_t4.8h
LIT0204	5.74	3127.2	21.6	25.9	5.8075	0.0614	0.000976	literature	Literature_Brix_TDS_t6.6h
LIT0205	5.89	3195.1	19.0	13.1	5.3074	0.0652	0.000762	literature	Literature_Brix_TDS_t0.8h
LIT0206	5.72	3261.0	24.7	14.4	5.8143	0.0696	0.000973	literature	Literature_Brix_TDS_t5.5h
LIT0207	5.62	5437.2	23.4	13.6	16.4598	0.0772	0.000933	literature	Literature_Brix_TDS_t22.5h
LIT0208	5.81	3171.1	24.2	12.2	4.8313	0.0619	0.000909	literature	Literature_Brix_TDS_t2.8h
LIT0209	5.77	3079.7	18.6	15.5	5.9279	0.0597	0.00101	literature	Literature_Brix_TDS_t5.9h
LIT0210	5.61	5027.3	22.6	24.1	17.6033	0.0725	0.001147	literature	Literature_Brix_TDS_t22.0h
LIT0211	5.63	4745.4	29.7	19.5	16.2318	0.0765	0.00109	literature	Literature_Brix_TDS_t21.0h
LIT0212	5.72	3148.0	16.3	8.7	6.4094	0.0654	0.000907	literature	Literature_Brix_TDS_t8.1h
LIT0213	5.69	3371.6	16.4	15.4	8.6971	0.0745	0.000938	literature	Literature_Brix_TDS_t11.9h
LIT0214	5.63	3982.9	19.0	19.6	12.686	0.0726	0.001289	literature	Literature_Brix_TDS_t18.0h
LIT0215	5.76	2995.4	24.9	8.0	5.7635	0.0683	0.001102	literature	Literature_Brix_TDS_t5.5h
LIT0216	5.73	3362.4	27.4	8.0	8.1825	0.0718	0.000941	literature	Literature_Brix_TDS_t10.8h
LIT0217	5.58	5514.3	18.4	14.4	17.6295	0.0817	0.001091	literature	Literature_Brix_TDS_t23.3h
LIT0218	5.84	3313.9	27.6	10.5	5.8869	0.0636	0.000953	literature	Literature_Brix_TDS_t1.7h
LIT0219	5.54	4855.2	22.3	22.7	16.2218	0.0738	0.001256	literature	Literature_Brix_TDS_t21.2h
LIT0220	5.74	4299.0	27.8	13.1	14.3258	0.0713	0.000888	literature	Literature_Brix_TDS_t19.1h
LIT0221	5.88	3117.7	25.6	10.1	5.3431	0.0582	0.000905	literature	Literature_Brix_TDS_t1.9h
LIT0222	5.91	3318.2	24.7	14.6	6.0896	0.0553	0.000888	literature	Literature_Brix_TDS_t7.0h
LIT0223	5.67	3740.6	17.2	20.8	10.1554	0.0686	0.001248	literature	Literature_Brix_TDS_t16.4h
LIT0224	5.8	3074.7	22.6	20.7	5.6023	0.0701	0.000974	literature	Literature_Brix_TDS_t5.7h
LIT0225	5.81	3155.6	17.0	16.3	8.9264	0.0548	0.000871	literature	Literature_Brix_TDS_t11.8h

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0226	5.63	3425.3	18.5	20.0	9.3076	0.0678	0.001097	literature	Literature_Brix_TDS_t11.8h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0227	5.72	2996.1	24.0	14.9	7.6698	0.0673	0.000915	literature	Literature_Brix_TDS_t1 0.4h
LIT0228	5.67	3643.4	26.6	9.3	9.9147	0.0743	0.001027	literature	Literature_Brix_TDS_t1 5.0h
LIT0229	5.54	3941.3	20.0	18.0	13.2873	0.08	0.001235	literature	Literature_Brix_TDS_t1 8.1h
LIT0230	5.59	5807.6	28.0	23.0	18.2293	0.0818	0.001234	literature	Literature_Brix_TDS_t2 3.8h
LIT0231	5.74	4601.2	29.8	17.8	15.7195	0.0718	0.000955	literature	Literature_Brix_TDS_t2 0.0h
LIT0232	5.78	3173.2	24.9	13.5	6.1077	0.0726	0.000878	literature	Literature_Brix_TDS_t7 .7h
LIT0233	5.68	3206.3	22.2	8.0	5.9959	0.0721	0.000921	literature	Literature_Brix_TDS_t7 .9h
LIT0234	5.86	3116.7	19.9	14.9	5.295	0.0641	0.000895	literature	Literature_Brix_TDS_t0 .5h
LIT0235	5.76	3240.7	22.2	8.0	5.4305	0.0731	0.001032	literature	Literature_Brix_TDS_t0 .5h
LIT0236	5.67	4940.7	23.7	20.1	16.4508	0.0759	0.000985	literature	Literature_Brix_TDS_t2 1.4h
LIT0237	5.56	3916.5	20.2	25.0	11.5009	0.0841	0.001103	literature	Literature_Brix_TDS_t1 7.4h
LIT0238	5.84	3131.4	22.1	13.4	5.6356	0.0587	0.000966	literature	Literature_Brix_TDS_t1 .1h
LIT0239	5.6	4527.1	24.4	9.9	14.9372	0.075	0.001109	literature	Literature_Brix_TDS_t1 9.5h
LIT0240	5.67	3191.9	29.0	19.9	8.3467	0.0746	0.001011	literature	Literature_Brix_TDS_t1 0.9h
LIT0241	5.63	5218.6	29.6	21.1	16.4681	0.0715	0.001145	literature	Literature_Brix_TDS_t2 2.2h
LIT0242	5.65	3319.4	21.8	16.7	4.5	0.0723	0.00095	literature	Literature_Brix_TDS_t1 .9h
LIT0243	5.9	2946.9	23.1	8.0	5.5845	0.0577	0.000869	literature	Literature_Brix_TDS_t1 .8h
LIT0244	5.73	2936.5	29.2	20.1	6.8981	0.0635	0.001078	literature	Literature_Brix_TDS_t8 .9h
LIT0245	5.71	3072.6	29.0	15.4	6.4009	0.0654	0.000781	literature	Literature_Brix_TDS_t9 .1h
LIT0246	5.76	3064.6	17.0	21.2	4.8208	0.0678	0.00097	literature	Literature_Brix_TDS_t3 .6h
LIT0247	5.7	3127.9	22.2	9.9	5.802	0.0693	0.000913	literature	Literature_Brix_TDS_t8 .0h
LIT0248	5.61	4871.2	25.6	21.0	16.0522	0.0763	0.001122	literature	Literature_Brix_TDS_t2 1.2h
LIT0249	5.72	3201.0	16.4	22.3	8.2182	0.0642	0.001012	literature	Literature_Brix_TDS_t1 0.6h
LIT0250	5.85	3356.4	21.5	10.6	5.402	0.0627	0.00093	literature	Literature_Brix_TDS_t2 .6h
LIT0251	5.66	3604.4	21.9	11.3	9.4596	0.0747	0.001096	literature	Literature_Brix_TDS_t1 4.2h
LIT0252	5.87	3203.9	27.0	11.8	5.4046	0.0596	0.001127	literature	Literature_Brix_TDS_t0 .5h
LIT0253	5.85	3239.4	20.7	16.8	4.894	0.0621	0.000861	literature	Literature_Brix_TDS_t1 .0h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0254	5.75	4421.0	22.6	18.7	15.8588	0.0746	0.000963	literature	Literature_Brix_TDS_t2 0.2h
LIT0255	5.75	3382.0	23.0	12.9	8.7933	0.0684	0.000944	literature	Literature_Brix_TDS_t1 2.1h
LIT0256	5.48	4772.2	21.5	17.0	15.3778	0.0881	0.001424	literature	Literature_Brix_TDS_t2 0.6h
LIT0257	5.82	3123.2	25.5	18.4	5.7653	0.0604	0.000974	literature	Literature_Brix_TDS_t7 .2h
LIT0258	5.72	3356.2	21.1	10.9	6.6249	0.0725	0.00099	literature	Literature_Brix_TDS_t8 .5h
LIT0259	5.55	5112.7	17.7	30.6	16.1083	0.0803	0.001294	literature	Literature_Brix_TDS_t2 1.6h
LIT0260	5.55	5235.6	29.9	26.4	16.42	0.08	0.00128	literature	Literature_Brix_TDS_t2 2.5h
LIT0261	5.71	3515.5	29.0	15.5	9.139	0.0686	0.000837	literature	Literature_Brix_TDS_t1 2.8h
LIT0262	5.71	3134.7	21.3	14.1	5.6292	0.0614	0.000877	literature	Literature_Brix_TDS_t5 .8h
LIT0263	5.84	3034.6	29.9	15.8	5.4047	0.063	0.000913	literature	Literature_Brix_TDS_t2 .2h
LIT0264	5.81	3129.0	28.5	17.8	6.1075	0.0652	0.001013	literature	Literature_Brix_TDS_t7 .5h
LIT0265	5.87	2890.5	26.8	8.0	5.6438	0.059	0.001012	literature	Literature_Brix_TDS_t7 .4h
LIT0266	5.59	4402.9	17.4	19.0	13.2369	0.078	0.001206	literature	Literature_Brix_TDS_t1 8.5h
LIT0267	5.72	3296.6	28.3	16.0	8.1374	0.0704	0.001012	literature	Literature_Brix_TDS_t1 0.8h
LIT0268	5.77	3004.6	23.0	15.4	5.9989	0.0757	0.000908	literature	Literature_Brix_TDS_t7 .9h
LIT0269	5.89	3207.9	27.0	16.6	7.8153	0.0575	0.000922	literature	Literature_Brix_TDS_t9 .9h
LIT0270	5.79	3417.7	19.7	14.9	9.455	0.0765	0.000982	literature	Literature_Brix_TDS_t1 2.6h

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0271	5.73	3915.2	16.9	18.3	11.5513	0.0691	0.001009	literature	Literature_Brix_TDS_t1 7.3h
LIT0272	5.86	3191.3	20.5	9.7	4.9796	0.065	0.00089	literature	Literature_Brix_TDS_t2 .8h
LIT0273	5.85	3309.3	28.9	18.2	5.1385	0.074	0.000813	literature	Literature_Brix_TDS_t3 .8h
LIT0274	5.55	5288.3	21.8	14.7	16.9967	0.0803	0.001259	literature	Literature_Brix_TDS_t2 2.7h
LIT0275	5.65	4034.5	20.7	18.6	11.4517	0.0659	0.00098	literature	Literature_Brix_TDS_t1 6.9h
LIT0276	5.63	4129.1	21.2	20.2	12.707	0.0771	0.001017	literature	Literature_Brix_TDS_t1 8.0h
LIT0277	5.65	3163.4	19.7	15.0	6.6527	0.0727	0.001004	literature	Literature_Brix_TDS_t8 .9h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0278	5.75	3412.6	29.5	15.1	9.1415	0.0661	0.000985	literature	Literature_Brix_TDS_t1 2.8h
LIT0279	5.89	3277.2	22.4	19.6	5.0242	0.0569	0.000951	literature	Literature_Brix_TDS_t1 .4h
LIT0280	5.57	4078.5	24.8	14.9	12.7658	0.078	0.001143	literature	Literature_Brix_TDS_t1 8.0h
LIT0281	5.67	4796.7	28.0	20.1	16.1178	0.0744	0.000943	literature	Literature_Brix_TDS_t2 0.9h
LIT0282	5.7	2996.0	28.4	14.4	6.6564	0.067	0.000939	literature	Literature_Brix_TDS_t8 .1h
LIT0283	5.66	5414.9	20.2	17.3	17.299	0.0663	0.000978	literature	Literature_Brix_TDS_t2 3.3h
LIT0284	5.84	3074.2	22.6	8.0	5.4226	0.0672	0.000855	literature	Literature_Brix_TDS_t0 .1h
LIT0285	5.75	3495.3	29.9	13.0	9.2166	0.0684	0.000934	literature	Literature_Brix_TDS_t1 2.5h
LIT0286	5.75	3195.6	22.9	13.0	6.6144	0.0697	0.000863	literature	Literature_Brix_TDS_t8 .5h
LIT0287	5.57	3199.2	18.9	22.8	7.5769	0.0758	0.001062	literature	Literature_Brix_TDS_t9 .7h
LIT0288	5.72	3673.9	22.0	18.1	9.4873	0.0724	0.001132	literature	Literature_Brix_TDS_t1 3.9h
LIT0289	5.78	3556.8	26.2	14.8	8.8534	0.0713	0.001024	literature	Literature_Brix_TDS_t1 3.6h
LIT0290	5.62	4667.0	16.8	15.3	15.6757	0.0642	0.001091	literature	Literature_Brix_TDS_t2 0.7h
LIT0291	5.74	3443.7	29.8	16.5	7.4998	0.0716	0.000972	literature	Literature_Brix_TDS_t1 0.2h
LIT0292	5.75	3332.7	19.7	17.4	8.6173	0.0768	0.000928	literature	Literature_Brix_TDS_t1 1.5h
LIT0293	5.74	3391.5	21.1	17.8	8.7603	0.06	0.001068	literature	Literature_Brix_TDS_t1 1.1h
LIT0294	5.72	3051.6	18.1	12.9	5.7512	0.0675	0.000957	literature	Literature_Brix_TDS_t3 .2h
LIT0295	5.65	5826.0	24.3	19.1	18.0999	0.065	0.00104	literature	Literature_Brix_TDS_t2 3.7h
LIT0296	5.66	3564.4	17.7	19.8	9.2317	0.0755	0.001094	literature	Literature_Brix_TDS_t1 4.3h
LIT0297	5.53	4020.7	23.8	19.3	12.9206	0.0848	0.001434	literature	Literature_Brix_TDS_t1 8.1h
LIT0298	5.71	3821.7	22.8	23.4	10.1533	0.0747	0.000922	literature	Literature_Brix_TDS_t1 6.4h
LIT0299	5.82	3047.7	26.7	18.9	5.4782	0.0629	0.000755	literature	Literature_Brix_TDS_t1 .6h
LIT0300	5.83	3184.5	27.6	15.7	4.9612	0.0577	0.000909	literature	Literature_Brix_TDS_t1 .6h
LIT0301	5.72	3735.5	22.1	17.4	10.7951	0.062	0.001006	literature	Literature_Brix_TDS_t1 6.7h
LIT0302	5.73	3032.8	16.7	12.9	5.7153	0.0665	0.000997	literature	Literature_Brix_TDS_t4 .9h
LIT0303	5.64	3805.7	24.7	19.6	10.2871	0.0703	0.000942	literature	Literature_Brix_TDS_t1 5.8h
LIT0304	5.75	2833.4	17.3	20.5	5.6954	0.0612	0.000835	literature	Literature_Brix_TDS_t6 .6h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0305	5.61	4385.9	22.2	9.3	15.7804	0.0684	0.001124	literature	Literature_Brix_TDS_t1 9.9h
LIT0306	5.83	3154.0	21.2	12.6	6.4757	0.0639	0.001096	literature	Literature_Brix_TDS_t8 .8h
LIT0307	5.83	3046.1	26.8	9.0	5.6769	0.0598	0.000972	literature	Literature_Brix_TDS_t2 .5h
LIT0308	5.87	3193.1	23.7	15.4	5.1074	0.0648	0.001158	literature	Literature_Brix_TDS_t1 .2h
LIT0309	5.89	3282.1	19.9	10.7	4.9413	0.0627	0.000884	literature	Literature_Brix_TDS_t0 .8h
LIT0310	5.68	3516.8	17.7	8.6	9.2258	0.0667	0.0009	literature	Literature_Brix_TDS_t1 2.8h
LIT0311	5.77	3071.9	18.3	18.5	5.0214	0.0698	0.000951	literature	Literature_Brix_TDS_t3 .5h
LIT0312	5.66	5154.5	27.6	19.4	16.5314	0.0649	0.001066	literature	Literature_Brix_TDS_t2 1.9h
LIT0313	5.62	4118.1	26.9	17.4	10.6883	0.0744	0.001101	literature	Literature_Brix_TDS_t1 6.6h
LIT0314	5.72	3486.9	29.6	13.3	9.3399	0.0725	0.000889	literature	Literature_Brix_TDS_t1 2.2h
LIT0315	5.58	3415.3	20.0	21.3	10.0336	0.0793	0.001209	literature	Literature_Brix_TDS_t1 3.0h

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0316	5.79	3402.6	26.5	17.1	8.5165	0.0696	0.000925	literature	Literature_Brix_TDS_t1 1.1h
LIT0317	5.89	3213.6	16.2	10.0	5.7211	0.0607	0.001065	literature	Literature_Brix_TDS_t1 .0h
LIT0318	5.82	3409.8	17.7	18.5	8.6722	0.0634	0.000806	literature	Literature_Brix_TDS_t1 3.1h
LIT0319	5.81	3168.6	20.7	10.4	5.4691	0.0695	0.000993	literature	Literature_Brix_TDS_t1 .8h
LIT0320	5.72	4203.2	28.8	11.7	13.3837	0.0721	0.001	literature	Literature_Brix_TDS_t1 8.4h
LIT0321	5.66	5376.7	26.5	17.7	17.4027	0.0716	0.001206	literature	Literature_Brix_TDS_t2 2.8h
LIT0322	5.79	2928.7	27.5	8.0	5.8726	0.0677	0.00092	literature	Literature_Brix_TDS_t5 .6h
LIT0323	5.82	3465.3	28.6	16.1	9.1905	0.0698	0.001085	literature	Literature_Brix_TDS_t1 1.9h
LIT0324	5.64	5292.4	20.8	21.3	16.6478	0.0828	0.001126	literature	Literature_Brix_TDS_t2 2.4h
LIT0325	5.75	3287.1	22.0	12.1	9.0022	0.0732	0.00106	literature	Literature_Brix_TDS_t1 1.9h
LIT0326	5.82	3130.0	28.1	17.6	5.9339	0.0646	0.001063	literature	Literature_Brix_TDS_t4 .8h
LIT0327	5.78	4254.8	21.9	21.2	14.5772	0.06	0.000884	literature	Literature_Brix_TDS_t1 9.2h
LIT0328	5.68	5000.5	19.2	25.1	17.0046	0.0714	0.001022	literature	Literature_Brix_TDS_t2 2.2h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0329	5.78	3007.1	17.0	8.6	5.4238	0.0599	0.001016	literature	Literature_Brix_TDS_t5.9h
LIT0330	5.7	3013.9	23.0	19.3	7.9137	0.0678	0.001106	literature	Literature_Brix_TDS_t10.7h
LIT0331	5.78	3294.1	24.2	14.1	5.9057	0.072	0.001067	literature	Literature_Brix_TDS_t8.2h
LIT0332	5.62	3830.7	21.3	17.4	10.1415	0.0794	0.00121	literature	Literature_Brix_TDS_t16.1h
LIT0333	5.81	3270.1	23.5	11.9	5.0216	0.0611	0.000873	literature	Literature_Brix_TDS_t0.8h
LIT0334	5.7	3458.9	25.0	25.3	8.3765	0.0664	0.000813	literature	Literature_Brix_TDS_t11.2h
LIT0335	5.6	4709.9	16.2	15.1	15.6191	0.0754	0.001193	literature	Literature_Brix_TDS_t20.7h
LIT0336	5.68	3084.7	27.0	15.3	6.15	0.0704	0.000993	literature	Literature_Brix_TDS_t8.5h
LIT0337	5.94	3282.1	19.8	14.0	4.7786	0.059	0.000979	literature	Literature_Brix_TDS_t0.1h
LIT0338	5.7	3755.3	18.4	16.8	11.499	0.0747	0.000912	literature	Literature_Brix_TDS_t17.0h
LIT0339	5.78	2927.3	19.2	8.0	5.6682	0.0611	0.000867	literature	Literature_Brix_TDS_t5.0h
LIT0340	5.69	5048.1	25.4	19.6	16.1032	0.073	0.000925	literature	Literature_Brix_TDS_t21.2h
LIT0341	5.9	3270.1	22.2	12.1	4.9124	0.0618	0.000721	literature	Literature_Brix_TDS_t2.4h
LIT0342	5.85	3145.4	26.6	8.7	5.4457	0.062	0.000955	literature	Literature_Brix_TDS_t3.9h
LIT0343	5.72	3246.8	24.9	19.2	5.8359	0.0627	0.000964	literature	Literature_Brix_TDS_t6.2h
LIT0344	5.67	3776.2	27.7	12.4	10.5363	0.0677	0.000924	literature	Literature_Brix_TDS_t16.5h
LIT0345	5.61	4600.4	25.5	17.8	15.5372	0.0852	0.001241	literature	Literature_Brix_TDS_t20.3h
LIT0346	5.73	3081.8	20.6	17.1	5.4106	0.0734	0.000912	literature	Literature_Brix_TDS_t1.5h
LIT0347	5.82	3233.5	23.8	12.8	5.3973	0.0689	0.000854	literature	Literature_Brix_TDS_t1.1h
LIT0348	5.67	5291.9	29.4	12.9	16.9657	0.0701	0.001093	literature	Literature_Brix_TDS_t22.2h
LIT0349	5.9	2915.4	22.3	16.8	5.4171	0.0639	0.000996	literature	Literature_Brix_TDS_t5.3h
LIT0350	5.85	3678.2	16.6	15.3	9.8017	0.0647	0.000869	literature	Literature_Brix_TDS_t15.4h
LIT0351	5.85	3186.5	16.9	14.3	5.5047	0.0626	0.000945	literature	Literature_Brix_TDS_t5.9h
LIT0352	5.74	3395.1	29.2	13.4	7.6064	0.0732	0.001004	literature	Literature_Brix_TDS_t10.4h
LIT0353	5.71	3266.1	21.8	19.2	7.3696	0.0738	0.000876	literature	Literature_Brix_TDS_t9.9h
LIT0354	5.5	5466.7	16.3	13.9	16.9987	0.079	0.001364	literature	Literature_Brix_TDS_t22.3h
LIT0355	5.72	3182.4	17.3	18.3	7.0367	0.0728	0.001017	literature	Literature_Brix_TDS_t9.6h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0356	5.73	3214.6	17.8	13.3	5.7701	0.0763	0.001104	literature	Literature_Brix_TDS_t5.3h
LIT0357	5.76	3521.0	18.9	12.9	10.5346	0.0697	0.000941	literature	Literature_Brix_TDS_t14.6h
LIT0358	5.62	5407.3	25.1	23.8	17.0494	0.0772	0.001174	literature	Literature_Brix_TDS_t22.3h
LIT0359	5.67	3435.3	25.4	20.4	9.041	0.0717	0.001138	literature	Literature_Brix_TDS_t13.6h
LIT0360	5.73	4565.6	16.1	19.3	15.9011	0.0677	0.000879	literature	Literature_Brix_TDS_t20.5h

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0361	5.89	3475.4	25.8	11.7	9.7602	0.0628	0.001072	literature	Literature_Brix_TDS_t14.2h
LIT0362	5.59	4657.7	17.2	11.5	16.4289	0.0772	0.000998	literature	Literature_Brix_TDS_t21.2h
LIT0363	5.82	3033.5	21.9	17.4	5.4794	0.0674	0.001041	literature	Literature_Brix_TDS_t5.3h
LIT0364	5.64	4888.7	28.8	17.7	16.2356	0.0707	0.001116	literature	Literature_Brix_TDS_t21.0h
LIT0365	5.77	3117.0	24.3	12.4	4.8033	0.0633	0.001015	literature	Literature_Brix_TDS_t0.3h
LIT0366	5.75	3237.5	25.1	14.4	5.3581	0.0728	0.000952	literature	Literature_Brix_TDS_t6.8h
LIT0367	5.76	4383.8	28.3	9.6	15.2348	0.069	0.001106	literature	Literature_Brix_TDS_t19.7h
LIT0368	5.75	3496.4	23.7	17.7	10.2783	0.0628	0.000898	literature	Literature_Brix_TDS_t14.6h
LIT0369	5.66	4548.0	22.6	17.3	15.4366	0.0698	0.001111	literature	Literature_Brix_TDS_t19.7h
LIT0370	5.66	3079.9	26.4	18.5	5.664	0.0712	0.000982	literature	Literature_Brix_TDS_t2.4h
LIT0371	5.8	3169.7	28.5	13.8	5.7751	0.0715	0.000984	literature	Literature_Brix_TDS_t2.7h
LIT0372	5.72	3837.2	19.3	16.4	11.8044	0.0597	0.00078	literature	Literature_Brix_TDS_t17.0h
LIT0373	5.72	3525.8	28.0	11.5	9.6103	0.0696	0.000974	literature	Literature_Brix_TDS_t13.7h
LIT0374	5.92	2988.7	25.9	20.6	5.2033	0.0599	0.001032	literature	Literature_Brix_TDS_t1.5h
LIT0375	5.7	3636.5	22.3	13.9	9.8025	0.0703	0.000923	literature	Literature_Brix_TDS_t15.3h
LIT0376	5.78	3310.8	27.0	8.7	5.7146	0.0648	0.000967	literature	Literature_Brix_TDS_t6.0h
LIT0377	5.83	3301.4	19.4	13.5	5.7244	0.0731	0.000911	literature	Literature_Brix_TDS_t5.2h
LIT0378	5.8	3471.4	16.2	20.3	6.9511	0.0683	0.001107	literature	Literature_Brix_TDS_t9.8h
LIT0379	5.51	4723.4	24.5	17.9	15.9435	0.0838	0.001546	literature	Literature_Brix_TDS_t20.9h

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
LIT0380	5.74	3792.2	18.9	14.8	9.8085	0.0726	0.00098	literature	Literature_Brix_TDS_t1 5.8h
LIT0381	5.56	4578.3	22.5	20.6	15.9364	0.0754	0.001222	literature	Literature_Brix_TDS_t2 1.0h
LIT0382	5.7	3211.0	29.4	15.2	5.5824	0.0701	0.001007	literature	Literature_Brix_TDS_t6 .2h
LIT0383	5.8	3231.5	20.2	14.3	6.1194	0.0622	0.000944	literature	Literature_Brix_TDS_t8 .0h
LIT0384	5.71	3134.8	19.8	14.1	8.0817	0.0722	0.000996	literature	Literature_Brix_TDS_t1 0.8h
LIT0385	5.71	3220.1	23.6	14.3	6.492	0.0683	0.000859	literature	Literature_Brix_TDS_t8 .0h
LIT0386	5.72	3131.5	26.1	16.2	7.5162	0.0793	0.000964	literature	Literature_Brix_TDS_t1 0.2h
LIT0387	5.68	4293.8	29.9	14.8	13.6394	0.0751	0.001019	literature	Literature_Brix_TDS_t1 8.6h
LIT0388	5.61	4239.5	16.0	11.0	12.5274	0.0703	0.001212	literature	Literature_Brix_TDS_t1 7.6h
LIT0389	5.77	3171.7	28.8	16.4	5.1703	0.0622	0.000878	literature	Literature_Brix_TDS_t4 .5h
LIT0390	5.77	3305.8	17.3	10.0	5.767	0.074	0.000859	literature	Literature_Brix_TDS_t3 .4h
LIT0391	5.64	5937.0	25.5	17.8	17.5691	0.0771	0.001168	literature	Literature_Brix_TDS_t2 3.7h
LIT0392	5.86	3289.7	22.9	8.8	5.0287	0.0633	0.001047	literature	Literature_Brix_TDS_t0 .1h
LIT0393	5.71	4087.9	18.7	17.2	12.2642	0.0667	0.00085	literature	Literature_Brix_TDS_t1 7.4h
LIT0394	5.72	3826.2	18.2	19.4	9.9522	0.0694	0.000983	literature	Literature_Brix_TDS_t1 5.5h
LIT0395	5.58	5230.3	16.1	16.8	16.8796	0.0751	0.001103	literature	Literature_Brix_TDS_t2 2.0h
LIT0396	5.88	3186.5	18.1	26.9	6.7036	0.0601	0.00097	literature	Literature_Brix_TDS_t8 .4h
LIT0397	5.76	3201.1	25.7	15.6	6.1221	0.0686	0.000877	literature	Literature_Brix_TDS_t7 .9h
LIT0398	5.75	3068.6	17.6	12.5	5.5376	0.062	0.000894	literature	Literature_Brix_TDS_t4 .7h
LIT0399	5.86	3226.8	20.5	14.4	5.589	0.0588	0.0009	literature	Literature_Brix_TDS_t3 .2h
LIT0400	5.82	3237.7	27.2	14.6	5.4838	0.0676	0.000809	literature	Literature_Brix_TDS_t7 .2h
S001	5.84	7031.8	18.9	26.8	4.53	0.002	0.0032	calibration	Natural
S002	5.3	5906.4	28.6	33.3	4.51	0.0014	0.0032	calibration	Natural
S003	5.62	5002.0	29.7	46.4	5.43	0.0045	0.0016	calibration	Natural
S004	5.73	7595.4	22.1	37.4	3.3	0.0022	0.0026	calibration	Natural
S005	5.91	7866.4	28.0	29.1	3.78	0.004	0.0024	calibration	Natural

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
S006	5.01	6878.8	27.4	26.0	5.0	0.0	0.0022	calibration	Natural
S007	5.87	4975.6	28.1	23.0	3.57	0.0028	0.0013	calibration	Natural
S008	5.97	7212.7	16.2	27.9	3.96	0.0025	0.0038	calibration	Natural
S009	5.11	6205.1	23.2	38.3	5.44	0.0027	0.0039	calibration	Natural

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
S010	5.6	6350.5	23.9	27.8	4.15	0.0029	0.0015	calibration	Natural
S011	5.19	4746.9	24.8	34.5	4.43	0.0004	0.0032	calibration	Natural
S012	5.88	7693.5	28.5	43.7	5.77	0.0027	0.0019	calibration	Natural
S013	5.71	5102.5	27.7	42.5	5.69	0.0029	0.0038	calibration	Natural
S014	5.58	5802.3	29.9	36.4	5.75	0.004	0.0008	calibration	Natural
S015	5.61	5945.8	27.7	35.2	3.73	0.0037	0.0009	calibration	Natural
S016	5.08	6951.6	23.6	18.3	3.92	0.0028	0.0026	calibration	Natural
S017	5.69	7878.7	27.2	33.5	5.29	0.0025	0.0005	calibration	Natural
S018	5.86	4553.3	20.4	40.6	3.86	0.0019	0.0022	calibration	Natural
S019	5.32	7238.7	25.4	12.5	4.9	0.0036	0.0016	calibration	Natural
S020	5.46	5318.0	16.7	20.5	3.74	0.0027	0.0034	calibration	Natural
S021	5.5	7185.6	19.7	24.9	5.87	0.0	0.0037	calibration	Natural
S022	5.04	7952.0	29.9	20.2	3.69	0.0024	0.0029	calibration	Natural
S023	5.56	7479.6	16.2	45.5	5.93	0.0044	0.0033	calibration	Natural
S024	5.82	6646.8	29.8	22.3	4.03	0.0007	0.0017	calibration	Natural
S025	5.62	5435.3	23.9	48.0	5.23	0.002	0.0011	calibration	Natural
S026	5.97	7647.7	29.1	41.9	5.32	0.0036	0.0038	calibration	Natural
S027	5.59	7952.5	15.5	11.1	5.9	0.0019	0.0007	calibration	Natural
S028	5.83	6842.3	21.5	45.1	4.0	0.0043	0.0038	calibration	Natural
S029	5.49	6716.2	20.9	14.7	3.21	0.0015	0.0012	calibration	Natural
S030	5.07	6192.4	20.5	46.7	5.2	0.0034	0.0009	calibration	Natural
S031	5.27	7397.3	25.8	22.0	4.45	0.0031	0.0006	calibration	Natural
S032	5.89	4506.0	19.6	21.3	5.38	0.0021	0.0008	calibration	Natural
S033	5.95	5961.9	18.9	23.7	3.13	0.004	0.0035	calibration	Natural
S034	5.78	6954.6	18.0	15.6	3.51	0.0046	0.0021	calibration	Natural
S035	5.12	7646.6	23.5	19.9	5.01	0.0029	0.002	calibration	Natural
S036	5.8	5523.1	20.5	20.4	4.55	0.004	0.004	calibration	Natural
S037	5.75	7068.3	28.1	13.3	4.09	0.0045	0.0012	calibration	Natural
S038	5.23	5887.6	25.0	21.9	3.3	0.0039	0.0031	calibration	Natural
S039	5.97	6605.7	17.6	41.4	5.74	0.0018	0.0028	calibration	Natural
S040	5.12	5196.1	26.3	19.4	5.47	0.0005	0.001	calibration	Natural
S041	5.33	5814.8	25.5	35.9	4.18	0.0034	0.0034	calibration	Natural
S042	5.27	5947.9	18.2	10.6	5.09	0.0026	0.0036	calibration	Natural
S043	5.13	7327.5	23.5	47.7	5.6	0.0023	0.0035	calibration	Natural
S044	5.82	6953.3	16.8	39.4	5.46	0.0029	0.0007	calibration	Natural
S045	4.88	4464.7	17.8	89.5	1.12	0.0157	0.0148	calibration	Artificial
S046	4.82	4564.8	19.8	36.1	1.21	0.0124	0.0122	calibration	Artificial
S047	5.19	5741.1	29.1	30.1	4.39	0.0259	0.011	calibration	Artificial
S048	4.59	4744.6	15.6	88.8	2.78	0.0156	0.0059	calibration	Artificial
S049	5.11	2711.3	24.0	52.7	2.19	0.029	0.0065	calibration	Artificial
S050	5.11	6787.3	18.6	46.7	2.7	0.0138	0.0014	calibration	Artificial

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sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
S051	5.13	2538.9	28.3	59.3	3.82	0.0136	0.0112	calibration	Artificial
S052	4.53	2635.3	26.1	72.0	4.04	0.0162	0.011	calibration	Artificial
S053	5.43	3705.7	18.1	49.6	4.5	0.0101	0.0035	calibration	Artificial
S054	4.78	3341.0	16.9	62.4	2.37	0.0182	0.0096	calibration	Artificial
S055	4.99	6956.8	28.7	92.6	4.22	0.0153	0.0091	calibration	Artificial
S056	5.11	3519.4	21.2	85.6	3.14	0.0225	0.0143	calibration	Artificial
S057	4.74	4666.0	28.2	56.8	4.14	0.0187	0.0149	calibration	Artificial
S058	4.55	5334.0	23.6	80.3	1.2	0.013	0.0014	calibration	Artificial
S059	5.45	3284.6	17.7	38.5	1.36	0.0195	0.0088	calibration	Artificial
S060	4.86	3295.4	18.7	26.6	2.35	0.0122	0.0076	calibration	Artificial
S061	4.59	3106.3	25.1	28.3	2.55	0.0167	0.0137	calibration	Artificial
S062	4.86	3297.5	25.3	35.0	4.0	0.0238	0.0193	calibration	Artificial
S063	5.15	2268.9	22.1	39.1	2.53	0.0175	0.0041	calibration	Artificial
S064	4.94	6014.3	16.8	35.7	4.29	0.0206	0.0177	calibration	Artificial
S065	4.72	5976.2	21.9	38.5	2.68	0.0196	0.0066	calibration	Artificial
S066	4.49	4721.2	25.1	43.2	4.54	0.0213	0.0128	calibration	Artificial

sample_id	pH	TDS	temperature	turbidity	sugar_pct	citric_acid_pct	ascorbic_acid_pct	source	notes
S067	5.13	3778.0	18.6	57.5	1.37	0.0159	0.008	calibration	Artificial
S068	5.26	3161.8	19.2	81.0	4.94	0.0124	0.0177	calibration	Artificial
S069	4.54	3282.9	23.7	62.1	2.58	0.012	0.0051	calibration	Artificial
S070	4.78	5776.1	23.9	92.7	1.14	0.0258	0.0061	calibration	Artificial
S071	4.84	4650.9	28.8	39.9	1.65	0.0183	0.0058	calibration	Artificial
S072	5.02	4869.9	23.0	70.2	2.64	0.0227	0.0081	calibration	Artificial
S073	5.28	5940.9	20.6	43.4	3.52	0.0131	0.0139	calibration	Artificial
S074	4.88	4955.3	25.0	31.2	2.42	0.0195	0.0083	calibration	Artificial
S075	4.98	5473.5	24.8	45.5	1.24	0.016	0.0149	calibration	Artificial
S076	4.55	5105.7	22.1	22.0	4.55	0.0102	0.0105	calibration	Artificial
S077	4.57	6335.5	26.1	74.9	3.68	0.0101	0.0008	calibration	Artificial
S078	5.12	6998.4	25.5	89.9	3.91	0.0145	0.015	calibration	Artificial
S079	4.79	2527.3	20.0	56.9	1.67	0.0184	0.0179	calibration	Artificial
S080	4.94	4236.5	22.9	76.7	1.52	0.0282	0.0089	calibration	Artificial
S081	5.29	3944.4	20.8	84.5	1.88	0.0139	0.0188	calibration	Artificial
S082	5.09	2249.0	18.5	51.1	1.34	0.0137	0.0011	calibration	Artificial

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